

## Article

# Abandoning the Car to Embrace the Bicycle in Urban France: A Model of Modal Shift for Daily Commuting Routines

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**Abstract:** The current climate context is prompting a reconsideration of transportation modes, with the bicycle offering significant benefits, including reduced emissions, noise pollution, and urban congestion and improved public health. Despite policies encouraging cycling, many urban French citizens struggle to adopt it. Rather than focusing on barriers, this study examines individuals who have reduced or abandoned car use in favor of cycling in urban areas. The study describes the modal shift process of 20 individuals from four cities in France, with diverse socio-economic backgrounds and ages. Through semi-structured, biographical interviews, key stages in the transition are identified: predispositions, triggering factors, exploration and trial, and long-term appropriation. The article also explores situations where cyclists revert to car use for specific trips. Findings reveal favorable conditions for cycling but also highlight challenges, offering insights that could influence urban planning policies in France.

**Keywords:** bicycle; sustainable mobility; modal shift; transport; city; ergonomics; sustainable transition



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## 1. Introduction

The current climate context requires the consideration of new, more sustainable forms of mobility. This is why, at the European level, territorial policies, as well as urban planning and mobility professionals, are reflecting on how to reduce the presence of automobiles in public spaces [1–4]. These strategies primarily aim to alter individual automobile travel practices by expanding modal shift options, promoting more ecological services (shared transport, bike or scooter-sharing schemes, carpooling, etc.), or improving existing infrastructure (e.g., reducing lanes in favor of public transport or bike lanes) [5].

A bicycle, in particular, offers multiple benefits [6,7]. Less polluting than the car at every stage of its lifecycle—from manufacturing and circulation to recycling—it is especially praised for not producing carbon dioxide (CO<sub>2</sub>), which contributes to climate change, or toxic air particles [8]. Being silent, it also reduces noise pollution in cities [9]. It also takes up less public space, for example, for parking or road infrastructure [3]. Furthermore, a bicycle is less expensive than a car, both in terms of its purchase price and maintenance costs [10,11]. It is also seen as beneficial for the local economy and tourism [12]. Finally, when used regularly, cycling is considered advantageous for one's health [13].

Despite these advantages, the car remains the primary mode of transport for daily trips in France, with 84.7% of domestic transport being carried out by private vehicles [14]. Cycling, on the other hand, remains marginal, and the French rank 19th out of 28 European

countries in terms of cycling as the primary mode of transport, far behind countries like Hungary, Denmark, or the Netherlands, where 36% of the population commutes by bike [15,16].

How can we explain the difficulty for the French, despite the policies and measures implemented, to adopt cycling? Rather than focusing on the factors inhibiting cycling practices, we have chosen to center our study on citizens who have decided to limit or even abandon car use in favor of cycling in urban and suburban areas. How did they manage to adopt this mode of transport in their daily routines, contrary to the general trend? What factors motivated or triggered this choice? What difficulties did they face, and what strategies did they employ to overcome them?

To answer these questions, we investigate the various aspects of the modal shift to cycling among 20 cyclists living in four French cities: Tours, Paris, Lille, and Nantes. This sample of 20 individuals allows for an in-depth exploration of their personal experiences, enriching our understanding of the motivations and challenges associated with adopting cycling. Each of the four selected cities presents distinct urban characteristics, varying mobility policies, and different cycling infrastructures, offering a wide range of experiences and contexts to explore.

After reviewing the existing studies on automobile and cycling mobility practices in France and identifying gaps in the literature, we present our data and qualitative approach. Regarding the results, we focus on the stages of modal shift towards cycling. A second section explains why even cyclists from urban areas do not completely abandon the car. Finally, we discuss these findings before concluding the paper in the final section.

## 2. Related Works

### 2.1. Automobile Mobility Practices in France

The driving license, beyond being an administrative document, carries a significant symbolic dimension and represents, through access to automobilism, a rite of passage into adulthood [2,17]. Automobile mobility then becomes the predominant mode of transport for individuals, notably because it is perceived as a way to comply with dominant social norms [18,19]. A car is also associated with a range of widely shared positive aspects. For example, it offers great flexibility in terms of travel times and, thereby, greater autonomy compared to other modes of transport that are bound to specific schedules [20]. In terms of practicality, a car is also linked to the ability to juggle activities that intersect work and daily life (e.g., picking up children from school, driving them to extracurricular activities, sports, shopping, etc.) and, thus, facilitates daily household needs [21]. Parenthood, in particular, is one of the events most likely to lead to the purchase of a car [22–24].

Beyond social norms and the perceived advantages of car use, automobilism remains the dominant mode of transport today, as urban planning policies continue to structure cities—residential areas, jobs, and commercial offerings—around car usage [2,25–27]. As a result, the automobile system is part of a vast sociotechnical system in which infrastructures, regulations, and technologies lead the population to rely heavily on this mode of transport [5,28,29]. Kaufmann et al. [29] already showed, in 2001, that the French population does not exclusively aspire to car use. It is more than ever being questioned by the population due to high costs, the occupation of urban space, and its ecological impact [18,30–32].

Studies highlight certain factors that hinder the shift to active mobility modes, such as cycling, and which include the lack of safe lanes for pedestrians and cyclists [33], private vehicle use habits [34], age and gender [35], or even race [36]. These studies, based on large-scale surveys, provide an overview of the drivers and barriers to making a modal shift. Other works focus on the life conditions that prompt individuals to consider new modes of transport.

## 2.2. Modal Shift Studied Through Mobility Biographies

It has long been acknowledged that daily mobility corresponds to a sequence of activities and movements that are indivisible and constrain individuals to a “program of activities” [37]. Therefore, the choice of a specific mode of transport must be understood by considering the entirety of this sequence, as well as within the context of life events.

Over the past thirty years, the concept of mobility biographies has developed, with various empirical analyses using both qualitative and quantitative methods. Beige and Axhausen [38], for example, adopted a longitudinal perspective to understand mobility trajectories through personal and family histories, places of residence, education, employment, and the possession of mobility tools (vehicle, transport cards, etc.). More recently, Jain et al. [39], using a stated-choice method, demonstrated how life events can radically alter household transport demand. They highlighted, for instance, how the birth of a child increases the likelihood of purchasing an additional car while reducing the likelihood of buying an electric bike or joining a car-sharing system. On the other hand, Rau and Matern [40] showed how sudden disruptions from significant life events and more gradual changes related to societal and environmental shifts led to changes in citizens’ relationships with the car.

Within the framework of mobility biographies, many studies have focused more specifically on cycling. Over ten years ago, Bonham and Wilson [41] conducted a qualitative study on the cycling experiences of Australian women throughout their lives, focusing on the circumstances under which they began or abandoned cycling. Van Acker et al. [42], for their part, analyzed the influence of childhood experiences on cycling use in adulthood, thus calling for the documentation of individuals’ “mobility capital” [43].

In France, Cailly et al. [44] and Adam et al. [45] have explored the socialization processes that lead to the adoption of cycling as a daily mode of transport in urban contexts. Their work details the importance of primary socialization for the later adoption of daily cycling practices, as well as the role of peer groups, which help develop additional sensory–motor and practical skills, such as orientation, dealing with motorized traffic, and dressing appropriately.

## 2.3. Research Gap

While previous studies provide valuable insights into the life trajectories associated with cycling, they often lack a systematic framework to structure and guide the analysis of the process of appropriation of cycling practices. For instance, research conducted in France by Cailly et al. [44] and Adam et al. [45] has highlighted the importance of primary socialization and peer groups in the adoption of cycling. However, these studies primarily focus on social and cultural influences without fully exploring the mechanisms of appropriation that transform cycling from a simple mode of transport into a daily integrated practice. Without a unified theoretical framework, these studies fall short in explaining how different dynamics interact and evolve over time. This limits their ability to generate actionable recommendations for public policies or local initiatives, which require a holistic understanding of the cycling adoption process to be effective.

Moreover, the absence of a systematic model makes it difficult to pinpoint the key stages in the appropriation process and the levers that can facilitate cycling adoption. As a result, mobility policies often remain superficial, promoting cycling without addressing the deeper structural and behavioral challenges that hinder its integration into daily life.

One promising theoretical approach to address this gap is the instrumental approach, conceptualized by Rabardel [46] and refined by subsequent research across various fields [47–53]. This approach focuses on the process of appropriation of any technical device, with particular emphasis on instrumental genesis: the process by which individuals

familiarize themselves with and integrate a tool (whether material or symbolic) into their practices, adapting it to their specific needs.

The instrumental approach provides a useful framework for understanding how people develop skills and knowledge through interactions with material or conceptual tools. Two dimensions are central to this process:

- Instrumentalization: this phase involves the user's adaptation of the tool, learning how to use it to meet their specific goals.
- Instrumentation: this phase refers to the evolution of the tool itself, which can be modified by the user over time to better align with their evolving practices and contexts of use.

In the context of cycling, this framework allows us to explore the dynamic interaction between the individual and the bicycle, shedding light on how individuals adjust their cycling practices over time and how their cycling experience contributes to modal shift. It also allows for a deeper understanding of the skills developed through cycling, providing a clearer picture of the factors that facilitate the transition from car use to cycling.

This theoretical grounding is particularly valuable for analyzing modal shift in urban environments, especially from the perspectives of instrumentalization (how individuals adapt their cycling practice) and skill development (how cycling experience contributes to sustained modal shift). The application of this framework in mobility studies is innovative, as it enables us to examine not only the adoption of cycling but also the evolution of this practice and its integration into daily commuting routines.

Given these insights, this study focuses on individuals who have intentionally reduced or abandoned car use in favor of cycling in urban environments. Our aim is to understand how they adopted cycling as a primary mode of transport for their daily routines, including essential trips such as commuting to work, running errands, picking up children from school, and other regular activities. What factors motivated or triggered this shift? What obstacles did they encounter, and how did they overcome them? Finally, what conditions help sustain cycling as a regular part of their daily mobility?

More broadly, we aim to characterize the stages of transition that individuals experience when switching from car use to cycling, providing a more comprehensive framework for understanding the dynamics of modal shift.

### 3. Materials and Method: A Qualitative Approach

#### 3.1. Participants

Our analysis is based on a corpus of 20 semi-structured and biographical interviews, fully transcribed, conducted between March 2023 and June 2023. The interviews, each lasting between 1.5 and 2 h, were carried out with 9 men and 11 women aged between 30 and 70 years. All of the individuals interviewed reside in urban areas within French metropolitan cities with populations of over 130,000: Lille, Paris, Tours, and Nantes. The four areas studied differ in terms of their size, infrastructure, population, and density.

We recruited our 20 participants using a combined approach, through social media and posters placed in public spaces. This strategy allowed us to reach a broad audience. From the responses received, we selected 20 individuals to obtain diverse and representative perspectives in terms of profession, gender, and age. A table summarizes the participants' profiles (Appendix A). All the first names have been changed.

Analyzing modal shift to cycling in a sample of 20 people in urban environments offers several advantages that make this approach both relevant and enriching. First, working with a smaller sample allows for in-depth qualitative interviews. This facilitates a detailed exploration of the motivations, experiences, and challenges faced by each individual. Such

a qualitative approach provides a nuanced understanding of the factors that influence cycling adoption, which could be lost in a large-scale quantitative study.

Furthermore, our sample of 20 individuals represents a range of backgrounds, ages, professions, and lifestyles. This diversity enables the capture of a wide spectrum of perspectives and strategies related to cycling adoption, enriching the analysis and making it more representative of urban realities.

We chose to focus on urban environments to study cycling adoption because these areas have a high population density, which leads to a variety of transport modes and mobility behaviors. This allows us to observe how cycling can integrate into a larger transportation system, interacting with other modes such as cars, public transport, and walking. Additionally, the four selected areas have well-developed cycling infrastructures, such as dedicated bike lanes and bike-sharing stations. This provides an interesting research context for evaluating the impact of these infrastructures on cycling adoption and user behaviors.

### *3.2. Research Methods: Between Semi-Structured Interviews and Biographical Interviews*

The interview conducted with each participant consisted of two parts: (1) the first was based on a semi-structured interview, and (2) the second was based on a biographical interview. Participants were informed that they could refuse to answer certain questions and that the analyses would be anonymized.

For the first part, we developed—based on our research questions—a set of questions aimed at systematically collecting themes to document: experiences, significant memories, and representations related to the car; experiences, significant memories, and representations related to the bike; the chronology and choice of modal shift towards cycling; difficulties, advantages, and surprises encountered; the objects of activity related to cycling practice; and relationships to other modes of transport and to living spaces.

The modal shift towards cycling was mainly documented around daily commuting routines, including the home-to-work commute, as this is quantitatively the primary reason for travel in France [20]. However, the study also examined other daily trips, such as errands, school runs, and recreational outings. This broader focus allows for a more comprehensive understanding of how cycling is integrated into everyday life and highlights the factors that support its adoption across various types of daily trips.

However, we also discussed other daily trips (for example, trips to visit family occasionally). The interview remained relatively open, allowing the participant to express their thoughts and perspective on the situation fairly freely, while the researchers occasionally prompted further responses based on those same comments.

For the second part of the interview, the biographical interview approach was used. The following instruction was given to participants: “Can you tell me your story with the bicycle, from the first time you used it until now?” This type of interview focused on the sequence of actions from the perspective of the subject who is telling the story. The subject describes the key and significant events that make the unfolding of their experience intelligible. Demazière [54] explains that the production of the narrative works by selecting events and episodes because the interview time is limited.

During this part of the interview, prompts encouraged the participants to verbalize their actions, thoughts, and feelings [55], as well as to understand the logic and processes involved in specific mobility situations. Our goal was to identify the phases of change, the obstacles encountered, and the influence of peers.



### 3.3. Analysis Methods

The collected material was subject to a multi-thematic coding process, involving a grid created from a variety of themes [56]. Multi-thematic coding aims to understand the material while engaging in a back-and-forth process with theory: it is through this movement that themes emerge. This coding technique involves reading the entire material without focusing on specific themes at first. By allowing for constant questioning of the initial theoretical concepts, it helps avoid the risk of circularity [57].

In practice, each interview was carefully read two to three times to identify emerging themes, such as territory or equipment. Meaning units, which could be a single expression, a sentence, or a series of sentences, were grouped under these themes. This multi-thematic coding analysis sits at the intersection of grounded theory [58] and theoretical coding [59]. At the crossroads of these two methods, it enables researchers to move beyond pre-established conceptual frameworks, allowing them to “cut through reality” and, thus, manage the density of the data. This approach not only prevents researchers from becoming locked into a priori categories that might bias the reading of the data and prevent new insights but also encourages thinking beyond these concepts and, more broadly, the theoretical frameworks themselves. For this reason, it seems to us the most suitable method for qualitative research.

In the following section, we highlight several quotes to illustrate our analysis, all coming from the study’s interviews.

## 4. Results

The analysis of the interviews first reveals the mechanisms leading to a modal shift towards cycling. Presenting the shift from one mode of transportation to another as a transition, thus, highlights the journey that individuals take in abandoning car use for cycling, taking into account a complex reality made up of multiple constraints, values, and desires. These mechanisms, some of which were also identified by Vincent [20], are as follows: *predispositions to change*, *triggers*, *exploration/trials*, and *instrumental genesis*. They are presented successively in this section.

### 4.1. Stages of the Modal Shift to Cycling

#### 4.1.1. Predispositions to Change

Predisposition to change can be understood as a person’s prior ability to do something or act in a specific circumstance or domain. This ability is not innate: it results from a set of experiences that allow an individual to interpret a situation [60].

#### *Questioning the Car*

One of the first predispositions to choosing cycling as a mode of transport is the gradual rejection of the car. This rejection is spontaneously mentioned by all of the interviewees for various reasons, ranging from strong rejection to simple inconvenience. First, there is frustration, or even exhaustion, due to the nuisances of road congestion:

*“The traffic, a part of me just wanted to avoid that”.*

Mathéo, 33, Stage Manager, Tours

All participants mentioned a sense of weariness regarding traffic difficulties in cities like Nantes, Lille, Paris, and Tours, where many roads have been narrowed in recent years to provide more space for cyclists. The feelings of stress, frustration, and even anger in bad traffic conditions, especially during rush hour, are also mentioned by eight participants. In this regard, cycling and walking are seen as less restrictive modes of transport that make it easier to move around the city center.

*“It really ruins your life, spending so much time looking for a parking spot morning and night, honestly”.*

Chloé, 28, Communications Officer, Nantes

Parking issues are also cited as negative aspects of car use. Finding a parking spot can significantly extend the commute time, creating a strong feeling of time lost among most participants.

Car commuting time, in fact, is often associated with wasted time that offers little opportunity for other activities, except for those who enjoy listening to the radio or podcasts. In contrast, public transportation is viewed as offering time savings, conducive to reading, watching videos, working, and even, for Alicia, leisure.

For six participants, driving a car is also associated with feelings of stress or anxiety related to a sense of road insecurity and a fear of accidents.

*“I’m convinced that the car is an object of death. I get knots in my stomach when I drive”.*

Garance, 42, Social Sciences Teacher, Paris

The cost of car use also contributes to the desire to limit its use. While maintenance and insurance costs seem inherent to car ownership and are relatively tolerated, it is primarily the costs of fuel and parking that participants seek to avoid:

*“I used to park on the quays to avoid paying for parking, but there was never a spot, so I would park past the last spaces, far from the university, just to park for free”.*

Mathéo, 33, Stage Manager, Tours

These costs are seen as unfair surcharges, especially as they have consistently increased in recent years in France.

### ***Physical Health***

Cycling, from the participants’ perspective, requires optimal physical health, which means having a certain amount of muscle strength in the legs, balance, endurance, good vision, and breath.

While all 20 cyclists interviewed affirmed that cycling helps them stay in shape, they all noted that it is also primarily because they are in good health that they practice cycling.

In this context, questioning car use is also linked, for 11 participants, to a desire to counteract sedentary behavior. Active modes of transport, such as walking and cycling, are seen as more conducive to maintaining good physical health:

*“And if I start using a car again, I’ll be sitting down, getting sluggish. But when you walk, when you run to catch the bus, you move. I didn’t want to get sluggish and end up with a big butt”.*

Alicia, 54, Jewelry Designer, Paris

### ***Experience with Cycling and Its Representation***

All of the participants interviewed have some degree of experience with cycling, either extensive or limited, starting in childhood and typically acquired within the family. During this period, they developed a set of motor and cognitive skills: accelerating, braking, coexisting with other road users, and navigating space.

Some, like Lisa (31, Bank Employee, Tours) or Mathéo (33, Stage Manager, Tours), have only used cycling on rare occasions (such as visiting friends in their childhood town), while others, like Manon (33, Life and Earth Sciences Teacher, Tours) or Patrick (66, Emergency Doctor, Lille), grew up cycling:

*“[My sisters and I], we were lucky to live in a building with a big park around it, which gave us the opportunity to learn to ride a bike very quickly and safely”.*

Manon, 33, Life and Earth Sciences Teacher, Tours

While our sample is too small to analyze the link between childhood cycling experience and choosing this mode of transport in adulthood, it is worth noting that the participants who switched almost immediately from car to bike once the opportunity arose (retirement, reduced family responsibilities, relocation, etc.) shared the common experience of having had a very favorable opinion of cycling based on increased use in childhood or adolescence. On the other hand, those for whom the transition was more difficult had limited cycling experience and retained a relatively unfavorable view:

*“Before, cycling was linked to constraints. I only used it when I didn’t have the car. [...] As a kid, I always had crappy bikes because of lack of money. So, I didn’t find any enjoyment in it. [...] In middle school, I walked because my mom’s apartment was just nearby. It was convenient. I had a bike that I used a few times to go to my friends’ houses. I went to my dad’s place once a week, and he would take me by car. In high school, I took the bus, Line 12. Same for going to town. I also walked to my theater class on Friday evenings. Anyway, my bike got stolen, and I didn’t get another one right away”.*

Mathéo, 33, Stage Manager, Tours

The excerpt above clearly shows how Mathéo developed a negative view of cycling during his childhood (due to using “crappy bikes” and “no enjoyment”). As a result, he did not buy another bike until he was 30, when he lost his driving license. A negative preconception of this mode of transport seems to limit the willingness to experiment with it, thus delaying the possibility of forming a more positive opinion. In contrast, the more someone masters a means of transportation, the more likely they are to explore it further and develop new skills related to it. Mathéo, for example, only returned to cycling when forced by the inability to drive, at which point he began to slowly reshape his view of cycling:

*“Today, cycling is part of my transport choices; I think about it. There’s the pleasure of walking and cycling”.*

Mathéo, 33, Stage Manager, Tours

Additionally, four participants had used motorized two-wheelers (motorbikes and scooters) during their adolescence or early adulthood, a phase during which they abandoned cycling. These experiences may have also contributed to developing transferable skills for urban cycling. The comparison between the equipment used for cycling and motorcycling is noteworthy:

*“For us, the cycling gear is really like motorbike gear. On two wheels, we are really vulnerable”.*

Séverine, 38, Unemployed, Tours

Another participant, Lisa (31, Bank Employee, Tours), used a scooter for six months to commute to work before switching to cycling. This phase served as a kind of transition to two-wheeled cycling, a mode she had rarely used during her childhood, adolescence, and early adulthood, making motorized two-wheelers a first step toward urban cycling mobility.

### **Ecological Concerns**

While the analysis of our data shows that ecological values alone are not the driving force behind the modal shift to cycling, they remain a key predisposition in the process of changing practices. Ten participants reported that the ecological benefits of cycling contributed to instilling in them a desire for a change in mobility.

For some participants, ecological reasons challenge a well-established car habit or even a strong preference for driving:



*“I must say it’s frustrating because I really like the car, and if I weren’t aware of the harm it does to the planet, I’d use it much more often. In fact, I probably use it too much compared to others. But I always ask myself the question. In cold weather, I try to motivate myself to take the bike, even though it’s hard”.*

Fabien, 30, Translator, Tours

Here, Fabien’s testimony shows how the value system can shake up a deeply entrenched set of practices. Thus, like Fabien, eight participants mention a sense of guilt when using a car, a feeling of making a harmful and morally questionable choice:

*“It took me time to think about my mobility choices and mature them. Some realizations came later than I would have liked. I feel guilty. When I don’t have to use the car, for example, to visit my mother. I often go to the [Atlantes Shopping Center] for groceries. The bags are very heavy, so I tend to take the car. I allow myself to use the car for shopping”.*

Mathéo, 33, Property Manager, Tours

This testimony reveals the sense of responsibility and guilt Mathéo feels. Thus, questioning mobility practices gradually pulls the car out of routine, incorporated use. Here, using the car for shopping is an “authorization” he grants himself, a deviation from his desire for more ecological mobility.

### ***The Influence of Close Ones***

We also observe among several participants the influence of people who can serve as role models for imitation. A close person—a child, a partner, a neighbor, a colleague, or a teacher—using a bicycle can, thus, become a model to follow for individuals and facilitate the integration of a bicycle as a possible mode of transport. This is what Maryse (62, Retired, Tours) describes, who abandoned the use of her car to take up cycling under the influence of her son. Social circles, thus, represent privileged spaces for initiating a change in practices.

### ***The Territory***

#### ***Terrain and Climate***

The terrain of the area also plays a role in the projection of a bicycle as a possible mode of transport. Marcia (31, Unemployed, Tours), thus, states that it took her a long time to consider cycling to a twin city of hers due to the steep slope that must be climbed to get there. Uphill routes can indeed require significant physical effort, which may discourage some people, even though they might actually be capable of it.

The climate also plays an important role. Rain, snow, wind, storms, cold, and, to a lesser extent, extreme heat are all reasons mentioned for not using a bicycle, especially by those who are trying, today, to cycle more in their daily commute but whose practice is not yet fully ingrained in their habits. Yohann (48, Consultant, Lille) spontaneously states that rain has long been an obstacle to cycling. Thus, rain, cold, and even extreme heat influence bicycle use.

#### ***Infrastructure***

Infrastructure specifically designed for cycling naturally encourages the shift towards using bicycles. Bicycle lanes, in particular, promote the adoption of cycling, especially because the feeling of insecurity associated with road traffic is reduced. Lucie shares how discovering new bike lanes upon her arrival in Tours led her to consider cycling as a viable mode of transportation. Yannick (34, Social Studies Teacher, Tours) describes how urban planning shapes the potential for cycling to become a viable transport option:

*“When I was young, we lived in Mauritania. The streets were unpaved, and cycling wasn’t practical, so we didn’t have bikes. We moved to Paris in 9th grade, and that’s when I started cycling to school. I’ve hardly stopped since”.*

Yannick, 34, Social Studies Teacher, Tours

#### 4.1.2. Triggers

The predispositions outlined above, however, do not seem sufficient on their own to trigger a change in practices. In order for a break in routine to occur, these predispositions must combine with a specific, often unexpected event that disrupts mobility routines. Four types of events—or triggering elements—were identified among our participants: biographical events; temporary inability to drive; unavailability of public transport; and urban planning, social, and technical innovations.

##### **Biographical Events**

###### *Moving and New Territories*

Among our participants, 11 reported starting to use a bicycle after moving or changing jobs. This kind of event seems to be a key trigger for a modal shift because it leads individuals to become familiar with new territories. Three main factors related to the characteristics of the territory then come into play: the discovery of new infrastructures, the distance, and the availability of transport options. For example, Lisa (31, Bank Employee, Tours) explains how her move to a new job location made her reconsider her transport options:

*“I knew it would be a hassle to get to my new job by bus. There were bus changes, and I hate that. [...] On the bike, I was worried about being on the road. I need a lot of safety. But then I realized it wasn’t so bad, and I got used to going by bike”.*

This passage shows how Lisa rethinks her daily commute after discovering that the bike route is safe and accessible.

Fabien (30, Translator, Tours) also embraced the bicycle more fully after moving closer to the center of Tours. Having lived in Essonne, where using a car seemed essential due to long distances and the lack of nearby public transport, his move allowed him to significantly reduce his car usage. The shorter distances to various amenities, like gyms, cafes, and stores, became a gateway for adopting cycling.

The distance between one’s home and workplace heavily influences the choice of transport. For example, Yannick (34, Social Sciences Teacher, Tours) was forced to buy a car due to his work assignments being further away from Tours. In 2021, he was stationed in Nogent-le-Rotrou, about 150 km from home:

*“We moved to Tours in the summer of 2021, my wife continues to work in Paris by train. And I was assigned to Nogent-le-Rotrou, it’s an hour and forty minutes by car, even two hours. There’s no public transport. There’s a train, but with transfers in Le Mans or Paris. Given my schedule, it wasn’t feasible. With three kids already, I wanted to be home in the evening, so I knew I would have to buy a car”.*

In this case, the distance and lack of public transport led Yannick to opt for a car, but when he was later reassigned closer to Tours, he returned to using the bike.

###### *Changes in Schedules*

A change in one’s schedule that provides more free time can also encourage individuals to adopt cycling as an alternative mode of transport. With less time pressure, they can enjoy their trips, take breaks, and explore more pleasant routes. For example, Maryse (62, Retired, Tours), France (70, Retired, Tours), and Heidi (66, Retired, Tours) all took up cycling after retiring. For example, for Heidi, a former librarian at a high school in Tours, retirement

provided the opportunity to reorganize her daily routine, and she describes how the time gained and flexibility in her schedule enabled her to switch to cycling:

*“I thought that when I retired, I would at least use the car as a minimum. But the bike became my main mode of transport. I don’t really like the bus because it restricts my schedule, and the bike, for me, has always meant freedom”.*

Similarly, Mathéo (33, Stage Manager, Tours), who works intermittently in the film industry, reflects on how his more flexible work periods allowed him to rely on cycling:

*“After my first shoot, I started to manage my intermittent work better and stopped accepting any and every job. I allowed myself longer breaks, which meant I was home more often and not always in Paris. During these break periods, I used the bike 100%. . . But during the hectic weekends of filming, when I barely have time to return, I drive to optimize time”.*

For Mathéo, the availability of time during his breaks enabled him to cycle, while during intense work periods, he optimized his time by using the car.

#### *Decrease in Family Responsibilities*

Six participants mentioned abandoning daily cycling due to family responsibilities at certain points in their lives. As Heidi (66, Retired, Tours) explains:

*“When I had my first child, it wasn’t common to carry kids by bike. [. . .] It was too rushed, the schedules were impossible, so I couldn’t manage everything. I was starting to teach, and I had other things on my mind”.*

This excerpt illustrates the link between parenthood and car dependency (already discussed in Sections 2.1 and 2.2). When children are young, especially infants, parents often have strict schedules, which can make cycling less feasible. As children grow and the burden of childcare decreases, particularly for mothers, cycling becomes a more viable mode of transport. The home-to-work trip often bridges the professional and personal spheres, connecting a series of activities: dropping children off in the morning, picking them up in the evening, going grocery shopping, etc. This chain of activities often influences the choice of transport, with cars being preferred for the flexibility they offer.

#### *Temporary Inability to Drive*

As mentioned earlier, routines are disrupted when an unpredictable and restrictive event occurs. Two key events that may lead to a temporary inability to drive are vehicle failure or the driver’s incapacity. For example, Séverine (38, Unemployed, Tours) describes how her car accident led her to reconsider her transport options:

*“We’ve had the bike since January. We only had one vehicle. My partner had an accident, and we looked at the second-hand market, but the prices were very high, and there wasn’t much choice. We were debating whether to buy another car or not. Eventually, we decided to buy a solid bike to fit our daughter’s seat. The accident was the trigger”.*

In this case, the temporary unavailability of the family vehicle prompted them to consider alternative transport options.

The temporary inability to drive can also be due to a personal issue, such as an injury or illness. Mathéo, for instance, had to adapt when he lost his driving license due to a DUI:

*“I lost my license too. I started taking the bus and train. That’s when everything switched. I bought a bike. I got my license back, and the shoot ended, but I continued to move around by bike. I handled all my admin tasks on the bike. It was a turning point”.*

This personal constraint led Mathéo to integrate cycling into his life more fully.

Similarly, Thomas (34, Teacher, Lille) recounts how he temporarily relied on a bike after his car broke down:

*“I had a car in the garage for two weeks, and I had no choice but to use the bike”.*

### **Unavailability of Public Transport**

The unavailability of public transport—due to factors such as limited schedules, strikes, construction, or new, poorly adapted routes—can also trigger a shift towards cycling. The bicycle often becomes an accessible and flexible alternative. For example, during the COVID-19 crisis in 2019, Patrick (66, Emergency Doctor, Lille) began cycling the 16 km to his hospital to avoid exposure to the virus through public transport.

The COVID-19 pandemic caused a significant rise in cycling in France, as many people sought alternative, safer ways to get around due to the restrictions on movement and the fear of public transport. Additionally, government initiatives, such as the expansion of temporary bike lanes and financial incentives for buying bikes, contributed to this surge in cycling.

### **Introduction of New Possible Transport Modes**

#### *Shared Bikes*

Our analysis suggests that social innovations impact mobility practices. Shared bikes, for example, enabled five of our participants to start cycling. Lisa (31, Bank Employee, Tours) and Victor (31, Unemployed, Nantes) use a bike-sharing service to commute to work each day. Séverine also tried a free temporary electric bike-sharing service provided by her city, which encouraged her to eventually purchase her own.

Bike-sharing services address several challenges, such as the issue of maintenance and storage. Lisa, for instance, explains that when she first started working at 20, she could have biked to work but was discouraged by the inconvenience of carrying a bike in and out of her apartment. Shared bikes solve this issue, similar to public transport, by eliminating the need for storage, maintenance, and ownership costs.

#### *The Rise of Electric Bikes*

The rise of electric bicycles can significantly contribute to the daily adoption of cycling as a mode of transportation, as it opens up more possibilities compared to traditional bikes. Thanks to the motor assistance, climbing hills or covering longer distances becomes feasible. Electric assistance, thus, facilitates the spatial extension of cycling practices. For example, Maryse, like the other five participants who use electric bikes, can now reach new destinations thanks to the electric assistance of her bike:

*“At the start of my retirement, I resumed my activities at La Camusière, sports and activism. I went there up to four times a week. I couldn’t climb the hill, so I used to take my car. But it consumed a lot, and I felt guilty. So, I thought either I get a car that consumes less, or I buy an electric bike. And so, I bought an electric bike and switched to a supplier that provides non-nuclear energy. It was just to go to Saint-Avertin [a town located at the top of a hill, where La Camusière is]. And I have my regular bike for the rest of the city. I also have an allotment garden, and I go there with the electric bike because it’s far, and it’s exhausting”.*

Maryse, 62, Retired Nurse, Tours

Thus, trips that were previously made by car—trips she could not replace with a traditional bike due to the terrain or distance—are now being done on an electric bike.

Moreover, an electric bike allows for more comfortable uphill rides with less effort and, therefore, less sweat. This is an important point mentioned by several users of e-bikes: the electric assistance offers the advantage of arriving at work in a presentable state without needing to bring a change of clothes or anticipate the availability of special facilities (shower, changing rooms, etc.). Additionally, transporting children or grocery bags is easier, as the weight added to the bike is offset by the motor’s power.

### *The rise of cargo bikes*

The growth of cycling practices, spurred by the arrival of electric bikes, is even more pronounced with the development of electric cargo bikes. Thus, our four participants who use electric cargo bikes, all of whom are parents of young children, rely on this type of bike for their daily commutes. Thanks to the space provided by the integrated cargo trailer, it becomes possible to combine activities such as leaving work, running errands, and dropping children off at extracurricular activities.

#### 4.1.3. Exploration and Trials

Exploration and trials around cycling refer to the initial phase in which users test various equipment, configurations, and practices. It is an experimental process that allows individuals to discover the possibilities offered by cycling and evaluate the comfort, performance, and functionality of different gear and routes. This phase is often characterized by curiosity and a desire to adapt the practice based on individual needs and lived experiences.

It should be noted right away that this phase is not linear and can involve multiple back-and-forth adjustments. Cyclists may start by exploring different navigation apps, testing performance-tracking devices, and checking various weather websites while adjusting their choices based on their experiences. For instance, an app that seemed promising may not meet their needs in real-life conditions, prompting them to try another one. Similarly, incorporating new equipment may require numerous adjustments.

#### *The Choice of the Bike*

Among all participants, 10 had already owned a recreational bike before using it for daily commuting; 1 participant did not have a bike and bought one at the time; 9 participants already had a recreational bike but chose to purchase a second bike specifically adapted for daily commutes. Among these, six participants invested in an electric-assisted bike (including four cargo bikes); three participants purchased a new bike.

Here, acquiring a bike can either involve buying a second-hand one or actively searching for new products on the market. In all cases, individuals are looking for a bike that meets their own criteria in terms of its shape, handlebar height, saddle, and weight:

*“It’s important to have a bike that fits your body. For now, this one works fine. The handlebar height, the saddle, and the comfort with the pedals. I don’t like bikes where you’re hunched over like a racer. I prefer an upright position, like women’s bikes. I don’t feel safe on a hunched bike”.*

Heidi, 66, Retired, Tours

The phase of reflection and trials before purchasing becomes even more important when it comes to electric-assisted bikes, whose average price is around EUR 2000:

*“When it came time to buy, it was mostly my wife who took care of it. I was dealing with the car. And she decided she was going to choose the cargo bike so that it would be suitable for her and her needs. The goal was to fit three seats, one for each child. She’s quite small, so the seat needed to be low and the bike needed to be maneuverable because she’s small and not very heavy. She did her research on the Internet, checked reviews, and then we tested bikes, looking on sites like Le Bon Coin. We tested bikes at the bike shop”.*

Yannick, 34, Social Sciences Teacher, Tours

Finally, five participants used a bike-sharing service when they shifted to cycling. Among them, two were satisfied with the service and do not plan to purchase a bike; one invested in an electric-assisted bike after testing the service for several months.

Choosing this option allows individuals to avoid the material constraints (e.g., purchase and storage) while also testing the bike before deciding to invest in one, especially if

it is an electric-assisted bike. This testing phase, however, can lead to dissatisfaction and the abandonment of cycling altogether, as was the case for Patrick during his initial trials with bike sharing in Lille:

*“I had considered and experimented with biking—during times when I didn’t have a car—by using Vélib’. I saw that the stations allowed me to move around and reach the stations. It wasn’t as convenient; the bikes weren’t always available, the weather was an issue, and it took a lot of time. The convenience of being in a car, being able to do my shopping on the way home, doing sports afterward, and putting my gym bag in the car, those were things I missed”.*

Patrick, 66, Emergency Doctor, Lille

This process of instrumentalization, as discussed in Section 2.3, involves transformations to the artifact (in this case, the bike), which can be local or global, temporary or more permanent. A wide range of equipment today offers the possibility to adapt one’s bike. New functions are then assigned to the artifact by the user and “improve” what was originally conceived by the designers to meet the needs of cyclists based on their personal characteristics and situational needs. These additions can affect the entire bike: clip-on saddlebags for the rear rack, baskets on the handlebars, saddle cushions, mirrors to attach to the handlebars, and vertical grips. Among our participants, most of these additions remain permanently on the bike, thus creating new functional properties that expand the possible activities. For instance, several participants explained that they added panniers to extend their bike use to grocery shopping.

### ***The Choice of the Route***

The new mode of transport is then put to the test on the road to explore the route to work. These trials may be conducted with a third party, especially if the individual has not ridden a bike in several years or has never ridden in an urban environment. Exploring the route allows individuals to scout future commutes and evaluate road safety conditions:

*“I was with my friend Théa, and I suggested we try the route to my work, we were doing a lot of sports together at the time. On a bike, I’m afraid to be on the road next to cars. I need a lot of safety. And I realized it wasn’t that bad, and I got into the habit of going by bike two or three times a week”.*

Lisa, 31, Bank Employee, Tours

*“I did a route survey by bike with my husband to see how to get there, because he navigates better than I do. Thanks to COVID-19, a lot of bike lanes opened up, so it’s not bad. I’ve tried other options, but I ended up sticking with the original route, it was the best”.*

Garance, 42, Social Sciences Teacher, Paris

This last excerpt clearly illustrates the trial-and-error process involved in route reconnaissance. Individuals consider different options and select the best routes based on their own criteria: safety, speed, or even the beauty of the area. This last criterion, for example, is what leads Heidi (66, Retired, Tours) to choose the Loire river paths to reach downtown Tours.

### ***The Choice of Equipment***

The participants also gradually try and introduce a variety of equipment to maximize comfort, safety, and functionality. Equipment refers to the clothing and accessories chosen for a specific activity. Perceived as an external element to a bike, it is mainly seen as a way to adapt cycling to one’s needs and the different situations encountered. It first serves as a response to weather conditions:



*“I bought enough gear for rain and cold, pants, to be really equipped. I have a big visor to protect myself, all that”.*

Maryse, 62, Retired, Tours

Equipment is also a way to address road safety issues:

*“I’ve made adjustments by buying equipment to make it more practical and safe: a mirror, a yellow vest, a helmet, lighting. . . Because I’m very aware of the risks of cycling. I realize that bikes aren’t always visible”.*

Patrick, 66, Emergency Doctor, Lille

Finally, equipment also provides a solution for managing situations where appearance matters, such as in jobs in retail or education:

*“I would get dirty. . . and that annoyed me. So I had a spare shirt in my bag—shirt and pants, in case I needed them in the locker at the school. As a teacher, I need to dress fairly well and be presentable”.*

Yannick, 34, Social Sciences Teacher, Tours

For some participants, “equipment” means adapting their attire: for example, Maryse (62, Retired, Tours) chooses to wear shorts to let rain hit her bare legs, or Marcia (31, Unemployed, Tours) avoids wearing a long skirt to pedal more easily.

#### ***The Choice of Technology (Navigation and Weather)***

Cyclists also test various technological tools that enrich their experience and improve their safety. Among these tools, navigation apps allow cyclists to plan routes that are adapted to bike paths and traffic conditions. Additionally, performance-tracking devices, such as speed and distance counters, help cyclists measure their progress and can provide a fun aspect:

*“I tested several weather websites, and the best one is the agricultural weather. That way, I know the exact moment I can go out and bike my route”.*

Marc, 45, Unemployed, Tours

As Marc explains above, testing different weather sites allows cyclists to choose the best time to ride, avoiding bad weather.

#### **4.1.4. Instrumental Genesis: Towards a Cyclist Identity**

This phase of exploration and experimentation, dynamic and iterative, leads to significant modifications of equipment and practices, gradually evolving into a sustainable cycling routine that allows each cyclist to find the configuration that best suits them.

Moreover, this cycling practice is accompanied by deeper transformations that gradually invite individuals to build a cyclist identity, particularly through the development of skills, the creation of social spaces, the expansion of cycling to other uses, and, finally, the strengthening of the objects involved in the activity.

#### ***Development of Skills***

The process of appropriation of the equipment involves a dual movement in the activity of individuals, directed towards the bike, the equipment, and the route but also towards the individual themselves. Depending on the situation, each person develops their own system of internal resources. We identify five specific areas of competence:

Acquiring experience in traffic situations;

- Knowledge of the area being traveled;
- Physical abilities;
- Technical skills related to the bike;

- Self-confidence.

#### *Acquisition of Experience in Traffic Situations*

Cycling requires a number of skills related to the actual operation of the mode of transportation. The basics of cycling involve acquiring good balance and mastering the three primary actions of cycling: pedaling, turning, and braking. All of the participants learned to ride a bike as children, so none had to develop these basic skills at the time of the modal shift, as Garance:

*"I know that at the beginning, I did quite a few scouting trips, especially with my partner, to get used to it and understand how it worked".*

Garance, 42, Social Sciences Teacher, Paris

However, several of them had to learn how to navigate in urban environments, that is, in public spaces shared with other road users (drivers, cyclists, scooter riders, pedestrians, skateboarders. . .):

*"When you're on a bike, there's a way to position yourself on the road so the person behind understands that you're going to turn left, for example".*

Philippe, 52, Craftsman, Nantes

The main skills acquired in this context relate to the formal and informal rules of interaction with other vehicle drivers, such as signaling a turn by raising an arm.

#### *Knowledge of the Territory*

Another skill acquired, as revealed through the interviews, relates to knowledge of the area traveled. This competence develops through regular cycling trips, with routes being perfected over time. For example, Marcia (31, Unemployed, Tours), looking to save effort, prefers routes with gentler slopes, while Maryse adapts her route based on the bike lanes she has identified over time:

*"I don't take the gentle slope through the village because the street is narrow and cars go really fast. I prefer Beaugaillard, there's a bike lane".*

Maryse, 62, Retired, Tours

Knowledge of the territory also includes knowing how to navigate using maps and identifying the relevant resources for that purpose:

*"I added a phone holder to guide me; it's a really strong magnet to attach it to the handlebars. I use the Geovélo<sup>®</sup> app. I had to get used to riding by following the route".*

Garance, 42, Social Sciences Teacher, Paris

The duration of the commute is also a factor frequently considered by individuals when using a bike to travel within their territory. Most participants use a geolocation app to estimate the time it will take to cycle from point A to point B. However, they know that the duration also depends on factors such as their health, the equipment used, the condition of the road, the destination (and its parking facilities), the weather (wind, extreme heat, etc.), the time of day, traffic conditions, and the purpose of the trip. For example, Yannick says he adapts his pace or even gets off the bike if necessary:

*"During periods of extreme heat, I would arrive early enough in the morning so that it wouldn't be a problem. It was a concern of mine, but it turned out not to be because I adjusted my pace, like walking up the hill if I had a meeting or something".*

Yannick, 34, Social Sciences Teacher, Tours

These factors influence the average speed, but over time, cyclists become better at assessing the duration of their bike rides with greater precision. After regular trips, Marcia knows exactly how long it takes her to get to various points in the city center.

In addition, parking in the city can be a major barrier to cycling. Therefore, knowing where bike parking is located becomes a skill in itself. Cyclists must anticipate a place to park, whether it is a designated facility or urban fixtures like racks, poles, or barriers. However, such parking is not suitable for bulky cargo bikes. Several participants mention preferring places with dedicated bike parking spaces and ample room for bikes:

*“Now, I go more to certain shops than others because I can park my cargo bike there more easily”.*

Séverine, 38, Unemployed, Tours

Similarly, Marcia (31, Unemployed, Tours) has developed a set of criteria for choosing where to park her bike: a secure, ground-fixed point, in areas with foot traffic, and often with two locks.

#### *Physical Abilities*

Cycling is also perceived by the participants as a mode of transport that helps develop physical skills related to endurance, instant power, that is, efficiency in generating speed, as well as lung capacity.

Endurance plays a crucial role: cycling long distances strengthens the cardiovascular system and improves the efficiency of the heart and lungs. Over time, the participants noticed an increased ability to sustain prolonged physical effort without getting tired, which benefits not just cycling but other physical activities as well:

*“It’s clear that since I started cycling, I feel in better overall shape. It’s hard to explain, but I just feel better physically in general”.*

Marc, 45, Unemployed, Paris

Instantaneous power becomes key, too. The ability to quickly generate speed, for instance, in a sprint or when going uphill, requires well-developed musculature. Leg muscles, particularly the quadriceps, hamstrings, and calves, are strengthened and adapted to these demands. This power is also synonymous with efficiency, as the cyclist learns to optimize each pedal stroke to maximize speed with minimal effort.

Breathing is another fundamental skill that develops with regular cycling. Cyclists must learn how to manage their breathing, inhaling deeply to supply their muscles with oxygen and exhaling effectively to remove carbon dioxide. This breath control not only improves cycling performance but also helps with self-regulation in stressful situations or intense physical effort.

*“Climbing the hill was hard the first two or three times, and after that, you get used to it, you don’t think about it anymore”.*

Manon, 33, Life and Earth Sciences Teacher, Tours

#### *Technical Bike Skills*

Cycling also allows individuals to develop technical skills related to bike maintenance and repair:

*“I’ve been cycling for a long time, so I know how to fix small things, like a chain derailment or a flat tire. I’m not afraid of that. My wife gets frustrated, so I do it for her. I’m not a real handyman for bigger repairs, though. If it’s something more complicated, like changing brake cables, I go to the bike shop. I’d like to learn to do everything on my bike, but I haven’t seriously tried yet. Right now, I have a problem with the gears shifting poorly, and I don’t know how to fix that system, but I’ll have to learn one day. For now, I take it to the repair shop”.*

Yannick, 34, Social Sciences Teacher, Tours

Yannick describes how his technical skills have developed over time through cycling. He mentions routine repairs like fixing a chain derailment or flat tire, which he now handles without fear. This indicates not only a familiarity with the bike but also a proactive approach to minor repairs. His attitude suggests that cycling is not just a physical activity for him but also an area where he has developed some technical expertise.

It is worth noting that, like Yannick, most participants prefer to delegate more complex repairs or a full bike overhaul (e.g., before a long bike trip) to professionals. However, all express a desire to improve their technical skills, a positive attitude toward learning, and a wish to become more self-sufficient.

#### *Self-Confidence*

It also became apparent that over time, the participants gained confidence and dared to venture beyond their usual routes, traveling farther and farther. They grew more confident in their ability to navigate urban traffic (including in very busy areas like central Paris) and in their physical abilities.

*“Now, on my new route, it’s much less safe. I wouldn’t have taken it at the beginning. When I started, it was important for me to have bike lanes to feel comfortable”.*

Garance, 42, Social Sciences Teacher, Paris

In conclusion, the various skills acquired, particularly the confidence gained through regular cycling, feed into an iterative cycle that strengthens the subjects’ predispositions.

#### *Development of Socialization Spaces*

Finally, cycling can also have a social dimension, fostering encounters and exchanges. The creation of cycling groups or communities of enthusiasts encourages the sharing of experiences and ideas, sometimes even promoting collective innovation. These interactions can lead to initiatives aimed at promoting the use of bicycles in broader contexts, such as improved cycling infrastructure.

Participating in group bike rides or joining a cycling community strengthens the sense of belonging, mutual support, and sharing. For instance, Yannick (34, Social Sciences Teacher, Tours) and Séverine (38, Unemployed, Tours) both describe how cycling invites them to interact with other parents who transport their children by cargo bikes at school drop-off. Here, the group is formed around a shared commitment to ecological values.

In other contexts, socialization is built around a shared interest in nature and botany, accessible through cycling trips. This is what Heidi (66, Retired, Tours) describes, as she rides along the Loire river paths into the center of Tours. She encounters other cyclists and can exchange thoughts about the plants they see along the way. Thus, the socialization spaces that emerge go far beyond the issue of daily commuting.

Modal shift to cycling can also lead individuals to more concretely join an already-formed group focused on cycling practice:

*“Cyclingdaily sparks other uses. I joined the ‘Droit au Vélo’ association. There is a community aspect to it”.*

Patrick, 66, Emergency Doctor, Lille

Patrick has, thus, become involved with the association Droit au Vélo, which works to promote cycling as the preferred mode of transportation in the Nord-Pas-de-Calais region.

#### *Extension of Cycling to Other Uses*

Finally, we observe that the modal shift to cycling for commuting leads participants to adopt the bicycle for other forms of use in various aspects of daily life. Thus, while recreational cycling experiences often facilitate the transition to using the bicycle for daily

commuting, it is also important to highlight that cycling, initially focused exclusively on commuting, can also expand into leisure activities.

For example, Garance (42, Social Sciences Teacher, Paris), who purchased an electric bike for commuting to work, gradually began using the bike for other types of trips, such as meeting friends in Paris, taking her children to school, and doing errands. Similarly, Lisa (31, Bank Employee, Tours) started using her bike for leisure activities once she had become accustomed to using it for commuting between her home and place of work.

### *Objects of Activity That Sustain Cycling Practice*

Our study also reveals that emerging objects of activity (OAs) play a crucial role in the sustainability of cycling practice. These motives, although initially secondary, strengthen over time through practice, thus broadening the scope of initial motivations. They contribute to firmly embedding cycling into the daily routines of users, transforming it into a long-term commitment, both personal and collective.

It is also important to emphasize that these OAs are dynamic and overlapping. An individual may adopt one or more of these objects depending on the journey or specific context. Eight objects of activity (OAs) were identified:

- Pragmatic OAs (OAPra);
- Ecological OAs (OAEco);
- Sportive OAs (OASpo);
- Health-related OAs (OASan);
- Hedonistic OAs (OAHed);
- Safety-related OAs (OASec);
- Financial OAs (OAFin);
- Social OAs (OASoc).

These will be presented following a summary table of the OAs relative to each participant (Appendix B).

For the participants, the bicycle primarily addresses a pragmatic goal (OA Pragmatic), meaning it is seen as a practical solution. It is perceived as efficient because it allows for relatively fast, easy, and unconstrained travel from point A to point B. Compared to a car, a bicycle is described as simpler, with fewer requirements; there is no need to find parking, wait for a car to warm up in the winter, deal with traffic jams, keep up with car documents, anticipate technical inspections, or maintain the vehicle. Ultimately, a bicycle provides a mode of transportation that compensates for the loss of flexibility and autonomy associated with other forms of soft mobility, like carpooling or public transport. The bicycle is also mentioned in various interviews as a way to alleviate the stress associated with driving and to permanently dissipate the mental load related to owning a motorized vehicle.

Cycling also serves an ecological goal (OA Ecological) and can even be part of a broader citizen and ecological commitment, often expressed through involvement in associations. Several participants, whose ecological motivation predominates in their cycling practice, report being active in environmental organizations. As Vincent (2008) shows, these activist practices suggest that the environmental values emphasized by participants are not merely justifications for their cycling practices.

It should be noted, however, that while the rise of cycling in cities suggests an increase in environmental concerns, the ecological objective (OA Ecological) often takes a backseat to pragmatic, athletic, health, or hedonistic goals. Thus, except for a minority of activists who are particularly informed about environmental issues, the primary motivation for cycling is it being practical, sporty, or pleasurable:

*“If it really wasn’t practical for me to take the bike—say, if the journey was longer—I would without hesitation take the bus, even though I find it really unpleasant. The*

*secondary motivation is sport. The bike lets me do my exercise. And the last thing is ecology, even though that's a really important aspect for me".*

Lisa, 31, Bank Employee, Tours

Sport motivation (OA Sport) is also a major factor in the desire to cycle. It even appears, in some cases, as an additional motivation that encourages participants to use a bike:

*"Sometimes, I can really feel lazy. But just the thought that I'll get my exercise on the bike motivates me to take it. It's really a reason that pushes me".*

Marcia, 31, Unemployed, Tours

This motivation is less pronounced among electric bike (e-bike) users but still exists. Yannick (34, Social Sciences Teacher, Tours), for example, removes the battery from his e-bike for ecological reasons, but also to get some exercise. Garance, on her part, finds that using the e-bike, even without heavy effort, helps to strengthen her back muscles:

*"I think it's good to move regularly. Also, I have back problems, so I think it helps me to tone up a bit".*

Garance, 42, SES Teacher, Paris

Health motivation (OA Health) is also mentioned but remains distinct from the sporting reason. Indeed, for some participants, cycling is much more than just a physical activity. It represents a true philosophy of life, centered on self-care and overall well-being. In this context, cycling becomes a tool for preventing various diseases, strengthening the immune system, and promoting longevity. The cyclist who integrates cycling into their daily routine often does so with the goal of preserving themselves, reducing stress, and feeling better overall. Cycling becomes a way to release endorphins, the "happiness hormones", which help combat stress and depression. Furthermore, the regular pedal rhythm combined with the scenery can have a meditative effect, helping individuals to refocus and clear their thoughts.

We also note a hedonistic motivation (OA Hedonistic) in the use of a bicycle, in the sense of a practice linked to the search for pleasure and positive sensations. Cycling outdoors allows individuals to reconnect with nature, appreciate the surrounding landscape, take deep breaths, and enjoy the sunshine. For participants, simply changing the environment and leaving enclosed spaces contributes to improving their mood and reducing anxiety:

*"There's a moment of relaxation on the bike, something I didn't get even while reading. Sometimes, on public transport, I couldn't even take my book out because it was too crowded. The bike is more comfortable: it's a moment of breathing".*

Garance, 42, Social Sciences Teacher, Paris

Two female participants also mention safety motivation (OA Safety). The bicycle is then used, especially for evening outings, to face the risks of public space aggression. They emphasize the ability to escape quickly, unlike walking or using public transport, where one has to wait at a bus stop for several minutes.

*"In the evening, I always take my bike, even though I'd like to walk around the city. But the bike is a defense tool for me. I'm less bothered, I can escape quickly. That's the main reason I use the bike in the evening".*

Marcia, 31, Unemployed, Tours

In these extracts, the bike, beyond being just a mode of transport, is used as a defense tool, as it allows, through acceleration, to extricate oneself from uncomfortable (e.g., "being bothered") or even dangerous (e.g., "being followed") situations. Moreover, a bicycle can also be perceived as a safer mode of transport compared to a car. For at least two other

participants, driving is seen as a dangerous activity (due to the risk of road accidents), with a bicycle providing a safer alternative. For example, Garance (42, Social Sciences Teacher, Paris) finds driving to be a source of suffering, stress, and fear. Discovering the bike, after years of putting up with bus rides that she no longer tolerated, was a relief. For Manon (33, Life and Earth Sciences Teacher, Tours), driving is associated with danger when drivers are tired. Therefore, she prefers the bike when she is not feeling physically well:

*“For me, there’s no safety in the evening. The headlights blind me. Driving at night is really tiring, and I find it very dangerous. It’s never changed. Plus, I have a Twingo, so compared to an SUV... [laughs]... When I’m sick, I think it’s stupid to take my car because I won’t be as attentive”.*

Manon, 33, Life and Earth Sciences Teacher, Tours

Finally, financial motivation (OA Financial) is mentioned but always in relation to the costs incurred from car use in the city. Compared to a car, a bike does not require fuel costs or parking fees. This motivation is also mentioned in relation to public transport. For instance, Mathéo (33, Stage Manager, Tours) chooses to use a bike or walk instead of paying for a bus ride.

Social motivation (OA Social) also plays a key role in sustaining the practice of cycling by facilitating meetings and interactions between individuals. This social dimension is particularly noticeable among parents who take their children to school by bike. For them, the trip becomes an opportunity for socialization, both with their children and with other parents or neighbors. This social aspect also extends to those who join cycling communities, whether in organized clubs or informal groups.

Despite the reasons that push individuals to adopt cycling as their primary mode of transport, our study reveals that cyclists rarely completely abandon their cars. This is addressed in the final section of results.

## 4.2. Cyclists Do Not Completely Abandon the Car

### 4.2.1. Several Situations of Using a Car

The modal shift to a bicycle does not completely eliminate the use of a car. In some cases, specific events may lead individuals to occasionally maintain car usage or even adopt other complementary modes of transportation. The reasons include *distance, work-life management, the transportation of goods, climate, and safety*.

#### *Distance*

It is first observed that for most long-distance trips between home and the destination, the use of a bicycle is limited. This is the case, for example, for the trips made by Lisa (31, Bank Employee, Tours), Yannick (34, Social Sciences Teacher, Tours), Manon (33, Life and Earth Sciences Teacher, Tours), Marc (45, Unemployed, Tours), and Chloé (28, Communications Officer, Nantes) when they visit their parents living in another region. Similarly, trips related to travel are often long distances and take place outside the geographic area of daily life (other regions of France, abroad, to visit grandparents, friends, etc.). In this case, trains and cars are the preferred modes of transport (and airplanes for international trips) because they allow for covering large distances.

A car is, therefore, used to reach areas that are far from home and poorly served by public transport. Manon explains that she replaces her bike when visiting friends in areas far from Tours:

*“And when it’s not for work, most of the times I use the car is to visit friends in areas far from Touraine”.*

Manon, 33, Life and Earth Sciences Teacher, Tours



Similarly, Heidi (66, Retired, Tours)—who practices ceramics—regularly travels to see a sculptor 57 km away from her home. This trip—one of the last she makes by car—seems impossible to complete by bike (it would take an average of 6 h for a round trip).

Outside of family visits, vacations, and access to poorly served areas, two participants (both teachers) who have been cycling for several years mention specific periods in their professional careers when they had to buy a car to reach rural areas due to job relocations.

### *Managing Work Life*

Other participants mention specific periods in their lives when using a mode of transport other than a car became difficult or even impossible due to time constraints. Professions with fragmented hours and several hours of break time during the day (such as teachers, for example) highlight the difficulty of using a bike when the trip has to be made multiple times in a single day.

### *Transporting Goods*

On the other hand, it is observed that some leisure or work-related trips are still made by car for a number of participants: these trips often involve transporting heavy or bulky goods that are difficult to carry by bike. For example, in the case of Fabien (30, Translator, Tours) and Manon (33, Life and Earth Sciences Teacher, Tours), transporting musical instruments for band rehearsals often requires using a car.

Manon, to get to the middle school where she works, sometimes abandons her bike; her car, with its cabin and trunk, allows her to transport equipment related to her teaching activity:

*“This year, I’ll go by bike except when I need to bring specific equipment, plants, or biology materials. . .”.*

Manon, 33, Life and Earth Sciences Teacher, Tours

The purchase of heavy furniture, or even moving houses, almost always justifies using a car, either by using one’s own vehicle, renting a car, or borrowing one from a relative:

*“At the time, for moving or buying heavy furniture, I would borrow my parents’ Renault Espace. We moved in 2013, a year after I arrived. At that time, my wife didn’t have a car”.*

Yannick, 34, Social Sciences Teacher, Tours

However, it should be noted that users of cargo bikes manage this constraint, as they can transport heavy and bulky goods on their bikes.

### *Climate*

Adverse weather conditions for cycling are varied and can pose significant obstacles for cyclists, both in terms of comfort and safety. While 13 of our participants report being equipped to face bad weather, 7 state that they opt for another mode of transport when weather conditions deteriorate.

*“In cold weather, I try to motivate myself to ride my bike, even if it’s tough. But in the rain, I’m not there yet, unless it’s just a light drizzle”.*

Fabien, 30, Translator, Tours

Unfavorable weather conditions for cycling include extreme heat, rain, cold, fog, and snow. Heat makes cycling exhausting and risky, especially on long trips. High temperatures also cause cyclists to sweat, which can limit cycling, especially for the commute to work:

*“If it’s too hot, I can’t take the bike. I can’t show up at a client’s place all sweaty, smelling strong [from sweat]”.*

Philippe, 52, Craftsman, Nantes

Rain and snow make the roads slippery, increasing the risk of falls and reducing visibility. Cold weather can cause pain and frostbite, while fog also limits visibility, both for cyclists and motorists. In particular, cycling in heavy rain or snow is considered dangerous because the cyclist becomes difficult to see, increasing the risk of accidents.

### **Safety**

Finally, several participants abandon cycling if the route they need to take seems too dangerous. The fear of accidents discourages cycling. As previously mentioned, this may be related to bad weather but also to dense traffic on narrow roads. This is the case for Heidi, who says she avoids certain trips she perceives as too dangerous:

*“There are some trips I avoid. I have a friend who lives in Ville-aux-Dames, and I drive to her house. I don’t take the bike over the bridge to get there. Too risky”.*

Heidi, 66, Retired, Tours

The presence of specific infrastructure then has a direct influence on abandoning the bike and is considered a key factor by the majority of participants. Thus, congested, poorly designed, or poorly maintained bike lanes can lead to a modal shift back to the car.

#### **4.2.2. Modal Shift or Systemic Reorganization of Different Transportation Options?**

Ultimately, it appears that modal shift is not simply a matter of switching from one mode of transportation to another but rather involves a systemic reorganization of the different mobility options available. This broader perspective acknowledges that individuals’ transport choices are often interconnected and fall within a larger framework of mobility practices.

When a person decides to use a bike instead of a car for certain trips, it does not necessarily mean that they are permanently abandoning the car. On the contrary, it may indicate an intention to integrate the bike into a set of transportation solutions, depending on the specific needs of each situation. For example, a user may prefer a bike for daily trips in the city while still keeping a car for longer journeys or situations that require transporting heavy loads.

This reorganization of transport modes also involves an awareness of the advantages and disadvantages of each option. Users become more sensitive to environmental, economic, and health-related issues, prompting them to think about the optimal use of each mode of transport in their daily lives. Thus, modal shift becomes a process of adaptation and experimentation, where individuals constantly adjust their choices based on contexts, available infrastructure, and social dynamics.

## **5. Discussion and Implications**

Our work, like other studies, considers cycling as a part of a wider, multi-determined transport system. For instance, regarding predispositions, we confirm studies on the socialization of travelers (travel socialization studies), such as those by Baslington [61], Faure [62] or Cacciari [63], which state that early mobility experiences influence the transport modes chosen in adulthood.

This perspective views mobility as a socialization process in its own right (similar to other cultural aspects) and considers that individuals who regularly used a bike during childhood are more likely to adopt this mode of transport in adulthood due to the skills acquired and the sense of capability that comes with it [45,64]. It seems that fostering confidence in one’s ability to use this mode of transport from a young age, through training programs and encouragement to use bikes, could be an effective strategy. Rigal [2], for example, suggests introducing a bicycle driving license, similar to a car driving license, to establish the bike as a legitimate and fully recognized mode of transport.

Moreover, our work shows, like existing studies, that the improvement of infrastructure encourages bike usage [20,43,65]. Studies focus particularly on the creation of quality bike lanes to enhance safety, as well as, and more importantly, the perception of safety [6]. The development of bike parking facilities, showers, changing rooms, and lockers would also promote cycling to work [66,67], with the availability of parking being especially important [68].

It should also be noted that social and technical innovations could be key triggers to encourage the transition to cycling. Bike-sharing programs seem to be valuable tools that allow people to try biking without having to purchase or store a bike. Since the introduction of the first Vélib' bike-sharing program, urban cycling for daily commutes has increased, particularly in city centers with more than 100,000 inhabitants, where 60% of trips are under five kilometers [16].

Regarding these kinds of triggers, financial incentives could also play a role. Companies could make biking financially attractive by offering mileage allowances, subsidies for purchasing bikes, or providing company bikes [69,70].

The rise of electric-assisted bicycles (e-bikes) seems to open up new possibilities for cycling in areas that were previously considered inaccessible, for example, hilly or more distant areas. E-bikes are particularly popular in regions where cycling rates have been historically low and mainly among older French citizens who had stopped biking, as well as women [16]. E-bikes also allow families with young children to travel by cargo bike.

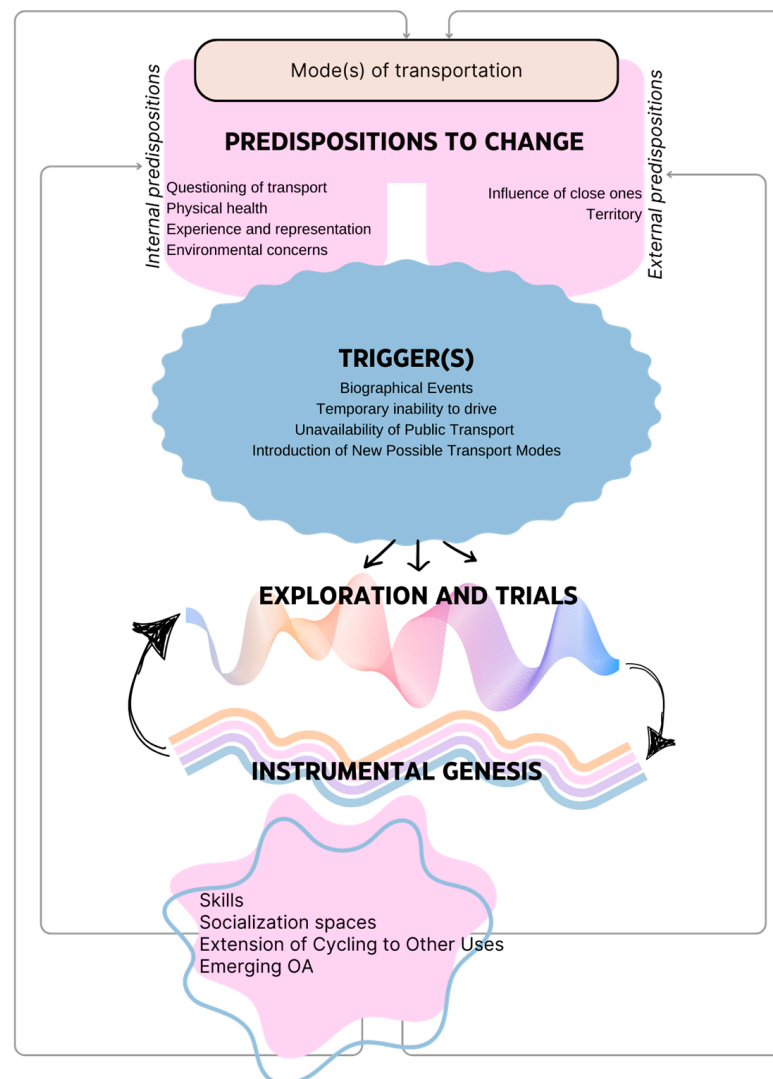
These studies provide valuable insights into the life trajectories related to cycling. However, there is still a lack of a systematic model to guide the analysis of the process of adopting cycling and, thus, assist in the development of concrete strategies in rapidly changing urban contexts.

Our study aimed to provide a detailed description of the process of appropriation that leads individuals to switch from a car to a bike. By adopting a fine-grained approach through semi-structured and biographical interviews, we were able to highlight the key stages of modal shift (Figure 1): predispositions to change, triggering factors, the exploration and trial phase, and the instrumental genesis.

The first two stages (predispositions to change and triggering factors) resonate with the work of Goodwin [70] and later with Vincent [20], which focus on the disruption of habits. Goodwin explains that a deeply ingrained habit can be broken if certain predispositions allow for a "slow erosion of the attractiveness of the habit" and if a "particular, sudden or unexpected" event occurs. The phases of predispositions and triggering factors are grounded in previous research, but here, they are validated and enriched by the participants' accounts. While these two stages are indeed the origin of the shift towards cycling, they do not alone explain the long-term adoption of cycling in daily transport practices. Our study, thus, proposes a model that extends from the initial situation (here, car use) to the integration of a new mode of transport into one's mobility practices (in this case, cycling).

An individual possesses internal predispositions (e.g., experience, physical health, etc.) and external predispositions (e.g., territory, etc.). The initial habit of a transport mode is disrupted by a triggering factor (e.g., moving houses, temporary inability to drive, new bike-sharing station, etc.). The individual then begins to consider the bicycle as a possible mode of transport and explores the necessary conditions for making this mobility a reality (e.g., type of bike, equipment, route, etc.). This exploration phase corresponds to the moment when the individual tests various equipment, configurations, and practices. It is an experimental process that allows for the discovery of the possibilities offered, evaluating the comfort, performance, and functionality of the different materials and routes. This phase is often characterized by curiosity and a desire to adapt the practice according to

individual needs and lived experiences. This phase is not linear and may involve multiple back-and-forth scenarios.



**Figure 1.** The key stages of modal shift.

This exploration and trial phase, dynamic and iterative, leads to significant modifications of the equipment and practices and transforms into a sustainable cycling routine, allowing each cyclist to find the configuration that works best for them. This phase resonates with the concept of instrumental genesis, explaining how individuals adapt their cycling practice to integrate it sustainably. Deeper transformations gradually encourage the individual to build a cycling identity, notably through the development of skills, the creation of socialization spaces, the extension of practice to other uses, and, finally, the reinforcement of activity objects.

These instrumental geneses will, in turn, feed back into the individual's predispositions, particularly by increasing their cycling experience. The practice of cycling can also influence external predispositions, as territorial policies may, for example, develop infrastructure to accommodate the growing number of cyclists [6].

This model ultimately allows us to understand mobility as a system subject to contingencies and variations. It further views the human not as a passive mobile individual but as an actor in their own transition, capable of taking initiatives, adapting, and modifying their actions to successfully achieve the objectives of the activity they pursue.

## 6. Conclusions and Future Research

The study aimed to understand how and why urban residents in France chose to limit or even abandon the car in favor of cycling. The main findings revealed the favorable conditions for modal shift towards cycling but also the difficulties encountered in certain situations.

From our perspective, another interesting result is that the modal shift is not merely a switch from one mode of transport to another but rather involves a systemic reorganization of the different mobility options available. This expanded perspective acknowledges that people's transportation choices are often interconnected and fit within a larger framework of travel practices. In future research, it could be relevant to develop a framework to understand this modal shift at the systemic level, such as the concept of "instrument system" [71,72]. This would highlight the complexity of the relationships between citizens and transport modes, shedding light on the process of adaptation and evolution that characterizes the use of instruments in daily practices.

This study also presents several limitations. First, the research was conducted with a limited sample of 20 participants. Although this sample allowed for rich and varied testimonies, it remains relatively small for drawing generalizable conclusions about the entire urban population. Cyclists' behaviors and motivations can vary significantly depending on numerous factors such as their age, socio-economic status, or geographic region. Therefore, some viewpoints or experiences may not have been sufficiently represented in the sample.

Furthermore, the diversity of the participants' profiles could also influence the results. While the sample includes individuals who chose to prioritize cycling for ecological, health, or economic reasons, perspectives from other groups might be lacking, such as those who continue to use the car for practical reasons or out of habit. This could introduce bias into the understanding of the motivations and challenges associated with modal transition. The marginalization of cycling practices based on gender, social, or ethnic origin is also clearly identified today and deserves further study in works related to cycling use [36,45,73–76].

Therefore, this study is mostly focused on motivational factors and favorable conditions for modal shift instead on focusing on obstacles or barriers. Our study aims primarily to understand the modal shift on an individual and subjective scale by identifying the driving forces towards modal shift. Nevertheless, it should be completed via the systematic identification of obstacles to mobility transitions.

Additionally, the urban context in which the study was conducted may limit the generalizability of the results. The dynamics of the transition to cycling can differ significantly between cities, depending on the availability of cycling infrastructure, local transport policies, and urban culture. Thus, the findings obtained in one city may not be applicable to other urban contexts or rural areas, where the factors influencing the decision to abandon the car may be radically different. For example, rural populations, especially those living in expansive areas with long distances, may face different challenges. In these areas, car use remains dominant, with four out of five trips being by car (Ministry of Ecological Transition and Territorial Cohesion, 2022). Cycling has also significantly declined in rural areas and multi-polarized communes, where cycling was once the most common mode of transport 25 years ago (5.8% in 1994) [16].

The findings of this study, while focused on urban contexts in France, can be applied to other regions, both domestically and internationally, with necessary adaptations for local specifics. The study emphasizes the importance of infrastructure, such as bike lanes and parking, which vary across cities. In areas with developed bike networks, like Paris, cycling adoption is easier, whereas in smaller or less populated cities, inadequate infrastructure can hinder the transition. Socio-economic factors also play a role, as access to cycling may be limited in rural or economically disadvantaged areas due to the cost of bikes. Tailored

financial assistance and local partnerships could help overcome these barriers. Additionally, cultural differences in transport habits may require region-specific awareness campaigns and infrastructure adjustments to promote cycling, especially in areas where cars dominate.

Internationally, the study's findings can be applied to cities with established cycling cultures, like those in Northern Europe (Amsterdam and Copenhagen) or Asia (Tokyo). However, developing countries, where cycling infrastructure is often lacking, could use these insights to gradually build secure cycling networks. Key areas for development include bike lanes, parking, and bike-sharing programs tailored to local climates and topographies. In countries like the US or Australia, where environmental concerns are growing, cycling adoption could be promoted through financial incentives, such as bike subsidies, and initiatives highlighting cycling's health and environmental benefits. Additionally, in hot, dry regions, climate-adapted solutions, like e-bikes and shaded bike lanes, are essential to make cycling more appealing despite extreme temperatures.

In summary, while this study offers valuable insights into the modal transition towards cycling, its limitations highlight the need for future research with a larger and more diverse sample, as well as the exploration of different urban and rural contexts, in order to better understand the complex dynamics that underlie this phenomenon.

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**Informed Consent Statement:** Informed consent was obtained from all subjects involved in the study.

**Data Availability Statement:** The data are not publicly available for privacy.

**Conflicts of Interest:** The authors declare no conflicts of interest.

## Appendix A. Profiles of the Participants of the Study

**Table A1.** Profiles of the participants of the study.

| First Name | Age | Gender | Profession | City   | Family               | Attitude Toward Cycling                                                        | Attitude Toward the Car                                                                                                    | Occasional Modes of Transport          |
|------------|-----|--------|------------|--------|----------------------|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| Fabien     | 30  | M      | Translator | Tours  | Lives with a partner | Personal bike (daily commutes: errands, outings, and leisure)                  | Personal car (family visits [monthly]; music rehearsals [weekly]; vacations [annually]; material transport [occasionally]) | Walking [occasionally]                 |
| Victor     | 31  | M      | Unemployed | Nantes | Lives alone          | Shared city bike service (daily commutes: errands, outings, and leisure; work) | Parents' car (visiting friends when returning to hometown) [about 6 times/year]                                            | Bus, train, and walking [occasionally] |

Table A1. Cont.

| First Name | Age | Gender | Profession                      | City  | Family                                  | Attitude Toward Cycling                                                                                                                                                                                               | Attitude Toward the Car                                                                                                                                            | Occasional Modes of Transport                                 |
|------------|-----|--------|---------------------------------|-------|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| Lisa       | 31  | F      | Bank employee                   | Tours | Lives with a partner                    | Shared city bike service (daily commutes: errands, outings, leisure; work)                                                                                                                                            | Partner's car (family visits [monthly]; occasional professional trips in the region [about 3 times/year]; vacations [annually]; material transport [occasionally]) |                                                               |
| Marcia     | 31  | F      | Unemployed                      | Tours | Lives with a partner                    | Personal bike (daily commutes: errands, outings, and leisure)                                                                                                                                                         | Borrowed car from friends (material transport [occasionally])                                                                                                      |                                                               |
| Lucie      | 32  | F      | Engineer                        | Tours |                                         | Electric-assisted cargo bike (daily commutes: errands, outings, and leisure; material transport; child transport [weekly])                                                                                            |                                                                                                                                                                    | Train (work [occasionally])                                   |
| Mathéo     | 33  | M      | Stage manager                   | Tours | Lives alone                             | Personal bike (daily commutes: errands, outings, and leisure)                                                                                                                                                         |                                                                                                                                                                    | Train (work [several weeks/year]); walking [occasionally]     |
| Manon      | 33  | F      | Life and earth sciences teacher | Tours | Lives alone                             | Personal bike (daily commutes: errands, outings, and leisure; work)                                                                                                                                                   | Carpooling with friends (family visits [monthly]; material transport [occasionally]; music rehearsals [weekly]; visits to friends [occasionally])                  |                                                               |
| Yannick    | 34  | M      | Social studies teacher          | Tours | Lives with a partner and young children | Electric-assisted cargo bike (daily commutes: errands, outings, and leisure; work; material transport [occasionally]; child transport [daily])<br>Personal bike (daily commutes: errands, outings, and leisure; work) | Personal car (family visits [monthly]; music rehearsals [weekly]; vacations [annually]; material transport [occasionally])                                         | Bus [occasionally]                                            |
| Séverine   | 38  | F      | Unemployed                      | Tours | Lives alone with young children         | Electric-assisted cargo bike (daily commutes: errands, outings, and leisure; work; material transport [occasionally]; child transport [daily])                                                                        | Parents' car (material transport [occasionally]; vacations [annually])                                                                                             |                                                               |
| Garance    | 42  | F      | Social studies teacher          | Paris | Lives with a partner and young children | Electric-assisted cargo bike (daily commutes: errands, outings, and leisure; work; material transport [occasionally])                                                                                                 |                                                                                                                                                                    | Bus, RER, metro, train, and walking [occasionally]            |
| Maryse     | 62  | F      | Retired                         | Tours | Lives alone                             | Electric bike (daily commutes: errands, outings, and leisure; associative missions)<br>Personal bike (daily commutes: errands, outings, and leisure; work)                                                            | Personal bike (for short trips)                                                                                                                                    | Borrowed car from friends (material transport [occasionally]) |

**Table A1.** *Cont.*

| First Name | Age | Gender | Profession             | City   | Family                                  | Attitude Toward Cycling                                                                                                                        | Attitude Toward the Car                                                                                       | Occasional Modes of Transport             |
|------------|-----|--------|------------------------|--------|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| Patrick    | 66  | M      | Emergency doctor       | Lille  | Lives with a partner                    | Personal bike (daily commutes: errands, outings, and leisure; work)                                                                            | Personal car (daily commutes: errands, outings, and leisure)                                                  | Metro [occasionally]                      |
| Heidi      | 66  | F      | Retired                | Tours  | Lives with a partner                    | Personal bike (daily commutes: errands, outings, and leisure; work)                                                                            | Personal car (material transport [occasionally]; visits to friends [occasionally])                            |                                           |
| France     | 70  | F      | Retired                | Tours  | Lives with a partner                    | Personal bike (daily commutes: errands, outings, and leisure; work)                                                                            | Personal car (material transport [occasionally]; leisure and outings [occasionally])                          |                                           |
| Alicia     | 54  | F      | Jewelry designer       | Paris  | Lives alone                             | Personal bike (daily commutes: errands, outings, and leisure; work)                                                                            |                                                                                                               | Metro (material transport [occasionally]) |
| Yohann     | 48  | M      | Consultant             | Lille  | Lives with a partner                    | Personal bike (daily commutes: errands, outings, and leisure; work)                                                                            | Personal car (material transport [occasionally]; leisure and outings [occasionally])                          |                                           |
| Thomas     | 34  | M      | Teacher                | Lille  | Lives with a partner                    | Personal bike (daily commutes: errands, outings, and leisure; work)                                                                            | Personal car (material transport [occasionally]; leisure and outings [occasionally])                          | Bus [occasionally]                        |
| Marc       | 45  | M      | Unemployed             | Paris  | Lives with a partner and young children | Electric-assisted cargo bike (daily commutes: errands, outings, and leisure; work; material transport [occasionally]; child transport [daily]) | Personal car (material transport [occasionally]; leisure and outings [occasionally])                          |                                           |
| Philippe   | 52  | M      | Craftsman              | Nantes | Lives alone                             | Shared city bike service (daily commutes: errands, outings, and leisure; work)                                                                 | Borrowed car from friends (material transport [occasionally]; leisure and outings [occasionally])             | Bus [occasionally]                        |
| Chloé      | 28  | F      | Communications officer | Nantes | Lives with a partner                    | Electric bike (daily commutes: errands, outings, and leisure; work)                                                                            | Personal car (family visits [monthly]; material transport [occasionally]; leisure and outings [occasionally]) | Bus and walking [occasionally]            |

## Appendix B. Synthesis of OAs for Each Participant

**Table A2.** Synthesis of OAs for each participant.

| First Name | Summary of Interview and Identified Activity Objects                                                                                                                                                                                                                                                                                                                                                                        |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fabien     | Since moving to Tours three years ago, Fabien uses a car much less, although he enjoys driving. He finds that cycling is as fast as driving into the city center (OAPra) and has the added advantage of being easy and free to park (OAFin). Currently, he tries to do most of his commuting by bike for ecological reasons (OAEco). He enjoys the sport it provides (OASpo) and has also taken up cycling tourism (OAHed). |
| Victor     | Victor has lived in the center of Nantes for several years. He regularly uses the city's shared bike service and also commutes by bus and walking. It is cheaper than using a car (OAFin) and more practical (OAPra). He feels that it benefits his mental health, especially since he suffers from frequent anxiety attacks (OASan).                                                                                       |



Table A2. Cont.

| First Name | Summary of Interview and Identified Activity Objects                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lisa       | After being a long-time car user, Lisa now uses a bike to get to work, except in bad weather. She finds it faster than public transport (OAPra). She also feels it benefits her physically (OASpo) and finds it more secure than other modes of transport (OASec), especially since she rides on dedicated bike lanes. The ecological aspect reinforces her choice (OAEco).                                                                                                                                                                                                                                                                                                                  |
| Marcia     | Marcia uses a bike for most of her daily travel. Unemployed, she chose cycling to stay active (OASan/OASpo) and save money (OAFin). She cycles for shopping, visiting friends, and exploring the city, enjoying the freedom it provides (OAHed). Marcia participates in local events and organized bike rides (OASoc), allowing her to meet other cycling enthusiasts and build a social network. Her commitment to cycling aligns with her desire to live more sustainably (OAEco). The bike is also a safer way to travel at night (OASec).                                                                                                                                                |
| Lucie      | Lucie lives in Tours with her husband and children but works in Paris. She commutes by train for work and uses a bike for her daily activities. She finds cycling more secure than driving (OASec) and, most importantly, more ecological (OAEco).                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Mathéo     | Mathéo, a freelance actor, lives in Tours but works mostly in Paris. He has been thinking about reducing his car use for ecological reasons (OAEco). It is also a financial gain, particularly compared to parking fees in his city, which have risen significantly (OAFin). It is also more practical since he no longer has to search for parking spaces (OAPra). He has bought a bike and started walking. He also frequently takes a bus. Cycling still presents some challenges for him (e.g., storing the bike in his building's basement). He still uses a car occasionally.                                                                                                          |
| Manon      | Manon, who lives in Tours, is a committed cyclist and enjoys cycling a lot (OAHed). She has used a bike for most of her daily commute for several years, initially for ecological reasons (OAEco)—it pollutes less and takes up less space in public areas—and for sports (OASpo). However, she had to buy a car two years ago due to being transferred to a work site in the northern part of the city and specific schedules (e.g., parent–teacher meetings). Overall, she finds the bike to be the least expensive form of transport (OAFin) and the fastest for getting around the city (OAPra). She also feels that a bike is safer, especially when she is too tired to drive (OASec). |
| Yannick    | Yannick, who has lived in Tours for several years (previously in Île-de-France), uses a bike extensively. He and his wife bought an electric-assisted cargo bike two years ago to drop their children off at school. It is easier to park (OAPra). His main motivation is ecological (OAEco), and he rarely uses the bike's battery to save energy. He also enjoys the physical workout the bike provides (OASpo). Yannick mentions that cycling has led to financial savings, although it is not his primary concern (OAFin). He discovered that cycling is a topic of conversation with other parents at school pick-up time, and he enjoys this (OASoc).                                  |
| Séverine   | After a car accident that resulted in material damage, Séverine and her partner decided not to replace their car. They bought an electric-assisted cargo bike instead, which was cheaper than repairing their car (OAFin). Séverine had already tried this type of bike through a city bike-sharing program and found it practical, especially for taking her daughter to school (OAPra). Séverine is very athletic, and cycling allows her to exercise daily, particularly since she is unemployed (OASpo). She also discovered that cycling is a topic of discussion with other parents at the school gates, which she enjoys (OASoc).                                                     |
| Garance    | Garance hates driving. She has sometimes been forced to use a car for professional reasons. She also used public transport a lot but did not like it. Since acquiring an electric-assisted bike a year ago, she really enjoys this new mode of transport, finding it the safest (OASec). She also considers it helps her maintain good general physical fitness (OASan) and especially strengthens her back (OASpo). The bike is also a form of relaxation for her (OAHed).                                                                                                                                                                                                                  |
| Maryse     | Maryse, now retired, uses an electric-assisted bike for most of her daily travels. Her main motivation is ecological (OAEco), although some trips, such as to Saint-Avertin (a town bordering Tours), were previously difficult due to steep roads. Since purchasing an electric bike, she finds it very practical for avoiding traffic (OAPra) and not having to search for parking in town. Maryse aims to maintain her physical fitness (OASpo) and enjoys being outdoors (OAHed). The bike is also a safer way to travel at night, such as when she goes to the cinema (OASec).                                                                                                          |
| Patrick    | Patrick, an emergency doctor at the Roubaix hospital, used to drive alone for years. For several years, he used a folding bike combined with public transport, mainly for ecological reasons (OAEco). Since the COVID-19 crisis in 2019, he now does the entire commute by bike. He is also involved in a cycling association (OASoc). He enjoys the sportiness of cycling (OASpo).                                                                                                                                                                                                                                                                                                          |

Table A2. Cont.

| First Name | Summary of Interview and Identified Activity Objects                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Heidi      | Heidi lives in Rochecorbon, 7 km from the center of Tours. Since retiring, she has made a greater effort to use a bike for her daily commute. She loves cycling on paths where she can observe nature (OAHed). She feels that it benefits her mind and keeps her in good spirits (OASan). The ecological aspect of cycling is important to her (OAEco). She also considers that it provides financial savings (OAFin) and saves time compared to driving, as there is no need to wait in traffic or search for parking (OAPra). |
| France     | France, a retired artist, uses a bike and bus for transportation. She still uses her car to transport art materials. She believes that cycling helps her stay active (OASan) while being practical (OAPra).                                                                                                                                                                                                                                                                                                                     |
| Alicia     | Alicia has completely switched from driving to cycling to stay in good physical shape (OASan) and avoid traffic (OAPra). She feels less stressed (OASan). She occasionally uses the RER (a commuter train) and metro for professional trips but has switched to cycling now that bike lanes are more widespread in Paris. This transition has helped her save time (OAPra) and discover the city from a new perspective (OAHed).                                                                                                |
| Yohann     | Yohann decided to reduce his car use in favor of cycling and public transport, which helps him avoid traffic jams (OAPra) and the stress associated with driving (OASan). Concerned about the environment (OAEco), he also wants to set an example for his child by adopting sustainable modes of transport. While he owns a car for family outings, he systematically prefers the bike and public transport for his daily commute.                                                                                             |
| Thomas     | Thomas uses a bike for commuting and leisure. A cycling enthusiast since childhood (OAHed), he enjoys the flexibility that a bike offers to navigate the city (OAPra). In addition to his daily commute, he regularly participates in bike outings with a cycling club (OASoc). For him, cycling is both a practical mode of transport and a way to socialize.                                                                                                                                                                  |
| Marc       | Marc uses an electric-assisted cargo bike for city trips. He enjoys the physical activity it provides (OASpo) while teaching his children the importance of sustainable transportation (OAEco). Although he owns a car for longer trips, he consciously makes an effort to use the bike whenever possible. He also uses the car to take his children to school, as it avoids parking difficulties (OAPra).                                                                                                                      |
| Philippe   | Philippe uses a bike for all his personal commuting. Cycling allows him to avoid traffic (OAPra) and stay active (OASpo). As a craftsman, he still uses a car for work-related travel,                                                                                                                                                                                                                                                                                                                                          |
| Chloé      | Chloé uses a bike as her primary mode of transport for daily commutes. She appreciates the speed and efficiency of cycling in a city with frequent traffic jams (OAPra). Concerned about the environment (OAEco), she decided to reduce her carbon footprint by prioritizing cycling and participating in events that raise awareness about sustainable mobility (OASoc).                                                                                                                                                       |

## References

1. Sucha, M.; Viktorova, L.; Risser, R. Can an Experience with No Car Use Change Future Mode Choice Behavior? *Sustainability* **2019**, *11*, 4698. [CrossRef]
2. Rigal, A. *Habitudes en Mouvement: Vers une vie Sans Voiture*; Metis Presses: Genève, Switzerland, 2020. [CrossRef]
3. Tortosa, G. L'aménagement d'un centre-ville sans voiture à Oslo: Véritable politique de restriction automobile ou simple requalification de l'espace public consécutif à l'effacement du trafic motorisé? *Flux* **2022**, *127*, 48–64. [CrossRef]
4. Wilkki, C.M.; Reeve, N. Communication from the Commission to the European Parliament, the Council, the European Economic and Social Committee and the Committee of the Regions on European Missions. European Commission. 2021. Available online: <https://ec.europa.eu/info/funding-tenders/opportunities/portal/screen/home> (accessed on 25 October 2024).
5. Cleuet, M.; Jehanno, A. Developing Sustainable Urban Mobility Plans—Guidelines for Mobilise Your-City Geographies. Available online: [https://www.mobiliseyourcity.net/sites/default/files/2023-03/SUMP%20Guidelines\\_final%20\(1\).pdf](https://www.mobiliseyourcity.net/sites/default/files/2023-03/SUMP%20Guidelines_final%20(1).pdf) (accessed on 25 October 2024).
6. Deuse, C. Perspectives Pour un Report Modal Vers les Vélos Classique et Electrique: Le cas des Déplacements Vers et au Départ de l'Université de Liège. Transport Studies. 2016. Available online: [https://matheo.uliege.be/bitstream/2268.2/1390/8/TFE\\_Deuse\\_Caroline\\_2015\\_2016.pdf](https://matheo.uliege.be/bitstream/2268.2/1390/8/TFE_Deuse_Caroline_2015_2016.pdf) (accessed on 1 November 2024).
7. World Health Organization; Regional Office for Europe. Walking and Cycling: Latest Evidence to Support Policy-Making and Practice. World Health Organization. 2022. Available online: <https://www.who.int/europe/publications/i/item/9789289057882> (accessed on 25 October 2024).
8. Gonzalez-Gorman, S.; Kwon, S.-W.; Patterson, D. Municipal efforts to reduce greenhouse gas emissions: Evidence from US cities on the US-Mexico border. *Sustainability* **2019**, *11*, 4763. [CrossRef]

9. Apparicio, P.; Gelb, J.; Jarry, V.; Lesage-Mann, É.; Debax, S. Exposition des cyclistes à la pollution sonore et atmosphérique à Lyon, France. *L'Espace Géographique* **2020**, *49*, 250–268. [CrossRef]
10. Brown, A.E. Car-less or car-free? Socioeconomic and mobility differences among zero-car households. *Transp. Policy* **2017**, *60*, 152–159. [CrossRef]
11. De Nervaux, L.; Guerra, L. Mobilités et Transition. 2023. Available online: [https://www.moreincommon.com/media/stkcinok/dc\\_mobilites-et-transition\\_final\\_compressed.pdf](https://www.moreincommon.com/media/stkcinok/dc_mobilites-et-transition_final_compressed.pdf) (accessed on 1 November 2024).
12. Middleton, J. *The Walkable City: Dimensions of Walking and Overlapping Walks of Life*; Taylor & Francis: London, UK, 2021. [CrossRef]
13. Papon, F.; Dusong, C. Bilan positif de la pratique du vélo-santé et environnement. In *Séminaire sur la Mobilité Soutenable Comparative Mexique-France*; Organised by el Cuerpo Academico de Economía Urbana y Regional de la Benemérita Universidad Autónoma de Puebla et le CONACYT (Consejo Nacional de Ciencia y Tecnología), 2016; 19p, Available online: <https://hal.science/hal-01455380/document> (accessed on 1 November 2024).
14. Ministère de la Transition Ecologique et de la Cohésion des Territoires. Bilan Annuel des Transports en. Données et Etudes Statistiques pour le Changement Climatique, L'énergie, L'environnement, le Logement, et les Transports. 2021. Available online: <https://www.statistiques.developpement-durable.gouv.fr/bilan-annuel-des-transports-en-2021> (accessed on 25 October 2024).
15. Razemon, O. 11 Chiffres sur le vélo en Europe. *Le Monde*. 11 May 2019. Available online: <https://www.lemonde.fr/blog/transports/2019/05/11/10-chiffres-sur-le-velo-en-europe/> (accessed on 25 October 2024).
16. Inddigo—Vertigolab. *Impact Economique et Potentiel de Développement des Usages du Vélo en France*; Dossier de la DGE, Government of France: Paris, France, 2020; p. 375. Available online: <https://preprod.charentestourisme.com/wp-content/uploads/2020/09/Impact-economique-et-potentiel-developpement-velo-en-france.pdf> (accessed on 25 October 2024).
17. Rérat, P.; Haldimann, L. Le permis de conduire chez les jeunes: Qui ne le passe pas (encore) et pourquoi? *Flux* **2020**, *119–120*, 5–24. [CrossRef]
18. Demoli, Y.; Lannoy, P. *Sociologie de L'automobile*; La Découverte: Paris, France, 2019. [CrossRef]
19. Cacciari, J.; Belton Chevallier, L. La démotorisation des ménages comme analyseur de la diversité des expériences de socialisation à la «norme automobile». *Flux* **2020**, *1*, 59–72. [CrossRef]
20. Vincent, S. Les «Altermobilités»: Analyse Sociologique D'usages de Déplacements Alternatifs à la Voiture Individuelle. Des Pratiques en Emergence? Ph.D. Thesis, Université René Descartes-Paris V, Paris, France, 2008. Available online: [https://consommations-et-societes.fr/uploads/uploads/documents/2008\\_theseSVINCENT.pdf](https://consommations-et-societes.fr/uploads/uploads/documents/2008_theseSVINCENT.pdf) (accessed on 1 November 2024).
21. Oakil, A.T.M.; Manting, D.; Nijland, H. Determinants of car ownership among young households in the Netherlands: The role of urbanisation and demographic and economic characteristics. *J. Transp. Geogr.* **2016**, *51*, 229–235. [CrossRef]
22. Clark, B.; Chatterjee, K.; Melia, S. Changes in level of household car ownership: The role of life events and spatial context. *Transportation* **2016**, *43*, 565–599. [CrossRef]
23. Lagrell, E.; Thulin, E.; Vilhelmson, B. Accessibility strategies beyond the private car: A study of voluntarily carless families with young children in Gothenburg. *J. Transp. Geogr.* **2018**, *72*, 218–227. [CrossRef]
24. McCarthy, L.; Delbosc, A.; Currie, G.; Molloy, A. Trajectories and transitions: Mobility after parenthood. *Transportation* **2021**, *48*, 239–256. [CrossRef]
25. Cervero, R.; Sarmiento, O.L.; Jacoby, E.; Gomez, L.F.; Neiman, A. Influences of built environments on walking and cycling: Lessons from Bogotá. *Int. J. Sustain. Transp.* **2009**, *3*, 203–226. [CrossRef]
26. Rocci, A. Changing Mobility Behaviors. Three Qualitative Case Studies: Paris, London and Nagoya. In Proceedings of the 11th World Conference on Transport Research, Berkeley, CA, USA, 24–28 June 2007; World Conference on Transport Research Society: Toulouse, France, 2017. Available online: <https://trid.trb.org/View/878218> (accessed on 13 April 2024).
27. Sheller, M.; Urry, J. The city and the car. *Int. J. Urban Reg. Res.* **2000**, *24*, 737–757. [CrossRef]
28. Spieser, K.; Treleaven, K.; Zhang, R.; Frazzoli, E.; Morton, D.; Pavone, M. Toward a systematic approach to the design and evaluation of automated mobility-on-demand systems: A case study in Singapore. In *Road Vehicle Automation*; Springer: Berlin, Germany, 2014; pp. 229–245. Available online: <https://dspace.mit.edu/bitstream/handle/1721.1/82904/main.pdf?sequence=1&isAllowed=y> (accessed on 25 October 2024).
29. Kaufmann, V.; Jemelin, C.; Guidez, J.-M. Automobile et modes de vie urbains: Quel degré de liberté? *Rech. Transp. Sécurité* **2001**, *72*, 92. [CrossRef]
30. Héran, F. La remise en cause du tout automobile. *Flux* **2020**, *119–120*, 90–101. [CrossRef]
31. Munch, E. Sustainable and slow cities: The quest for conviviality. *Sustain. Sci. Pract. Policy* **2024**, *20*, 2375810. [CrossRef]
32. Mwale, M.; Luke, R.; Pisa, N. Factors that affect travel behaviour in developing cities: A methodological review. *Transp. Res. Interdiscip. Perspect.* **2022**, *16*, 100683. [CrossRef]
33. Limanond, T.; Butsingkorn, T.; Chermkhunthod, C. Travel behavior of university students who live on campus: A case study of a rural university in Asia. *Transp. Policy* **2011**, *18*, 163–171. [CrossRef]
34. Van Veldhoven, Z.; Koninckx, T.; Sindayihebura, A.; Vanthienen, J. Investigating public intention to use shared mobility in Belgium through a survey. *Case Stud. Transp. Policy* **2022**, *10*, 472–484. [CrossRef]

35. Cortez-Ordoñez, A.; Tulcanaza-Prieto, A.B. The Effect of Gender and Age in Small Bicycle Sharing Systems: Case Study from Logroño, Spain. *Sustainability* **2023**, *15*, 7925. [CrossRef]
36. Archer, D.N. “White Men’s Roads through Black Men’s Homes”: Advancing Racial Equity through Highway Reconstruction. *Vand. L. Rev.* **2020**, *73*, 1259.
37. Jones, P.M. New approaches to understanding travel behaviour: The human activity approach. In *Behavioural Travel Modelling*; Routledge: London, UK, 1979; pp. 55–80. [CrossRef]
38. Beige, S.; Axhausen, K.W. Long-term and mid-term mobility decisions during the life course: Experiences with a retrospective survey. *IATSS Res.* **2008**, *32*, 16–33. [CrossRef]
39. Jain, T.; Johnson, M.; Rose, G. Exploring the process of travel behaviour change and mobility trajectories associated with car share adoption. *Travel Behav. Soc.* **2020**, *18*, 117–131. [CrossRef]
40. Rau, H.; Matern, A. Mobility practices in a changing climate: Understanding shifts in car ownership and use across the life course. *Mobilities* **2024**, *19*, 869–888. [CrossRef]
41. Bonham, J.; Wilson, A. Bicycling and the life course: The start-stop-start experiences of women cycling. *Int. J. Sustain. Transp.* **2012**, *6*, 195–213. [CrossRef]
42. Van Acker, V.; Mulley, C.; Ho, L. How childhood experiences affect travel behaviour differently across generations: An example of structural equation modelling in mobility biographies research. In *Mobility and Travel Behaviour Across the Life Course*; Edward Elgar Publishing: Cheltenham, UK, 2020; pp. 190–205. [CrossRef]
43. Rocci, A. De L’automobilité à la Multimodalité? Analyse Sociologique des Freins et Leviers au Changement de Comportements vers une Réduction de L’usage de la Voiture. Le cas de la Région Parisienne et Perspective Internationale. Ph.D. Thesis, Université René Descartes-Paris V, Paris, France, 2007. Available online: <https://theses.hal.science/tel-00194390> (accessed on 12 July 2024).
44. Cailly, L.; Huyghe, M.; Oppenchaim, N. Les trajectoires mobilitaires: Une notion clef pour penser et accompagner les changements de modes de déplacements? *Flux* **2020**, *3*, 52–66. [CrossRef]
45. Adam, M.; Ortat, N.; Poisson, A. Compétent-es, mais exposé-es: Du tricycle au vélo utilitaire, socialisations et trajectoires des cyclistes urbain-es. *Espaces Sociétés* **2022**, *184*, 33–49. [CrossRef]
46. Rabardel, P. *Les Hommes et les Technologies. Approche Cognitive des Instruments Contemporains*; Armand Colin: Paris, France, 1995. Available online: <https://hal.science/hal-01017462> (accessed on 12 July 2024).
47. Rabardel, P.; Beguin, P. Instrument mediated activity: From subject development to anthropocentric design. *Theor. Issues Ergon. Sci.* **2005**, *6*, 429–461. [CrossRef]
48. Rabardel, P.; Bourmaud, G. From computer to instrument system: A developmental perspective. *Interact. Comput.* **2003**, *15*, 665–691. [CrossRef]
49. Bationo-Tillon, A.; Poret, C.; Folcher, V. Appréhender le développement des organisations à la croisée du cours d’action et de l’approche instrumentale: La perspective transitionnelle. *Activités* **2020**, *17*, 30. [CrossRef]
50. Trouche, L. De L’approche Instrumentale à L’approche Documentaire du Didactique: Une Histoire D’appropriations Croisées et D’hybridations Théoriques. *Actes du Séminaire National de Didactique des Mathématiques*. 2020, p. 5. Available online: <https://hal.science/hal-02512596/> (accessed on 17 November 2024).
51. Albert, M.; Judon, N.; Folcher, V.; Bourmaud, G.; Pueyo, V.; Larbaigt, J.; Garrigou, A. Apport des approches instrumentale et ergotoxicologique pour la conception: Le cas du matériel de pulvérisation des pesticides en viticulture. *Le Trav. Hum.* **2021**, *84*, 197–232. [CrossRef]
52. Gras Gentiletti, M. Résumé de thèse. “Soutenir la dimension constructive de l’activité instrumentée par des dispositifs techniques à base d’intelligence artificielle: Une approche développementale de la conception centrée sur le modèle du sujet capable”. *Activités* **2023**, *20*, 10. [CrossRef]
53. Lahoual, D. Conceptualiser les Activités Constructives et le Développement du Sujet Capable. Le cas de la Médiation à L’art Orientée Jeune Public dans un Musée d’art Moderne et Contemporain. Doctoral Dissertation, Université Paris 8-École Doctorale “Cognition, Langage, Interaction” (ED 224), Paris, France, 2017. Available online: <https://hal.science/tel-01621677/> (accessed on 19 November 2024).
54. Demazière, D. L’entretien biographique et la saisie des interactions avec autrui. *Rech. Qual.* **2011**, *30*, 61–83. [CrossRef]
55. Cahour, B. Émotions, affects et confort comme nouveaux déterminants de l’activité et de l’usage. In *Ergonomie, Conception de Produits et Services Médiatisés*; Valléry, G., Ed.; Presses Universitaires de France: Paris, France, 2010; pp. 273–305. [CrossRef]
56. Ayache, M.; Dumez, H. Le codage dans la recherche qualitative une nouvelle perspective? *Le Libellio d’Aegis* **2011**, *7*, 33–46.
57. Dumez, H. *Méthodologie de la Recherche Qualitative: Les Questions clés de la Démarche Compréhensive*; Vuibert: Paris, France, 2016.
58. Glaser, B.G.; Strauss, A.L. *La Découverte de la Théorie Ancrée: Stratégies pour la Recherche Qualitative*; Armand Colin: Paris, France, 2010.
59. Böhm, A. Theoretical coding: Text analysis in grounded theory. In *A Companion to Qualitative Research*; Flick, U., von Kardorff, E., Steinke, I., Eds.; Sage: London, UK, 2004; pp. 270–275.
60. Lahire, B. *Portraits Sociologiques: Dispositions et Variations Individuelles*; Armand Colin: Paris, France, 2005.

61. Baslington, H. Travel socialization: A social theory of travel mode behavior. *Int. J. Sustain. Transp.* **2008**, *2*, 91–114. [CrossRef]
62. Faure, E.; Bochaton, A.; Demailly, K.-E. Construire un Système Vélo. 2024. Available online: <https://hal.science/hal-04707130/> (accessed on 1 September 2024).
63. Cacciari, J. Les Mobility Biographies Research et les Travel Socialization Studies Sous le Regard de la Sociologie de la Socialisation: Vers L'examen de la Production Sociale des Choix de Mobilité Spatiale. Available online: <https://hal.science/hal-03165953/> (accessed on 3 November 2024).
64. Flamm, M. Se passer de la voiture demande beaucoup de savoir-faire. *La Rev. Durable* **2005**, *18*, 31–34.
65. Roetynck, A. PRESTO Cycling Policy Guide. Electric Bicycles. 2010. Available online: [https://www.rupprecht-consult.eu/fileadmin/migratedRupprechtAssets/Documents/PRESTO\\_Cycling\\_Policy\\_Guide\\_Electric\\_Bicycle.pdf](https://www.rupprecht-consult.eu/fileadmin/migratedRupprechtAssets/Documents/PRESTO_Cycling_Policy_Guide_Electric_Bicycle.pdf) (accessed on 1 November 2024).
66. Balsas, C.J. Sustainable transportation planning on college campuses. *Transp. Policy* **2003**, *10*, 35–49. [CrossRef]
67. Akar, G.; Clifton, K.J. Influence of individual perceptions and bicycle infrastructure on decision to bike. *Transp. Res. Rec.* **2009**, *2140*, 165–172. [CrossRef]
68. Degand, J.-C. Le stationnement, composante essentielle d'une politique pour le vélo. *Transp. Urbains* **2020**, *2*, 24–28. [CrossRef]
69. Vandenbulcke, G.; Dujardin, C.; Thomas, I.; de Geus, B.; Degraeuwe, B.; Meeusen, R.; Panis, L.I. Cycle commuting in Belgium: Spatial determinants and 're-cycling' strategies. *Transp. Res. Part A Policy Pract.* **2011**, *45*, 118–137. [CrossRef]
70. Goodwin, P. Changes in transport users' motivations for modal choice: Passenger transport. In *Report of the 68th Round Table on Transport Economics*; European Conference of Ministers of Transport: Paris, France, 1985. [CrossRef]
71. Bourmaud, G. Les Systèmes D'instruments: Méthodes D'analyse et Perspectives de Conception. Ph.D. Thesis, Université Paris VIII Vincennes-Saint Denis, Paris, France, 2006. Available online: <https://theses.hal.science/tel-00109046> (accessed on 9 September 2024).
72. Bourmaud, G. L'organisation Systémique des Instruments: Méthodes D'analyse, Propriétés et Perspectives de Conception Ouvertes. Colloque de L'association pour la Recherche Cognitive-ARCo'07: Cognition-Complexité-Collectif. 2007, pp. 61–76. Available online: <https://inria.hal.science/inria-00191128/document> (accessed on 8 November 2024).
73. Abord de Chatillon, M.; Ortat, N.; Sayagh, D. Le vélo: Un objet qui révèle, renforce et perturbe l'ordre du genre. *Rech. Sociol. Anthropol.* **2021**, *52*, 25–51. [CrossRef]
74. Aldred, R.; Watson, T.; Lovelace, R.; Woodcock, J. Barriers to investing in cycling: Stakeholder views from England. *Transp. Res. Part A Policy Pract.* **2019**, *128*, 149–159. [CrossRef] [PubMed]
75. Mundler, M.; Rérat, P. Le vélo Comme outil D'empowerment. Les Impacts des Cours de Vélo pour Adultes sur les Pratiques Socio-Spatiales. Les Cahiers Scientifiques du Transport-Scientific Papers in Transportation. 2018, p. 73. Available online: [https://afitl.msh-lse.fr/tl\\_files/documents/CST/N73/Rerat73.pdf](https://afitl.msh-lse.fr/tl_files/documents/CST/N73/Rerat73.pdf) (accessed on 11 October 2024).
76. Sayagh, D. Construction Sociospatiale de Capabilités Sexuées aux Pratiques Urbaines du vélo. Les Annales de la Recherche Urbaine 2017. Available online: <https://www.annalesdelarechercheurbaine.fr/IMG/pdf/aru112-12.pdf> (accessed on 12 November 2024).

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